

Heterogeneity of radial spokes structural components and associated enzymes in *Tetrahymena* cilia

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Abstract

Radial spokes (RS), the T-shaped, multiprotein complexes of motile cilia, transmit regulatory signals from the central apparatus to the outer doublet complexes, including inner dynein arms. In the vast majority of ciliated species, RSs assemble as repeats of triplets (RS1-RS2-RS3), and each spoke is associated with a different subset of inner dynein arms. Studies in *Chlamydomonas* and mice sperm flagella led to the identification of RS proteins (RSPs) and revealed that some structural components are either RS1- or RS2-specific. In contrast, the protein composition of RS3 remains largely unknown. We used the ciliate *Tetrahymena thermophila* to investigate the protein composition of individual RSs, including the poorly characterized RS3. The *Tetrahymena* genome encodes three RSP3 paralogs. Using engineered RSP3 knock-out mutants and previously studied RS mutants with *CFAP61*, *CFAP91*, or *CFAP206* deletion and complementary approaches, including bioinformatics, total ciliome comparisons, and cryo-electron tomography with subtomogram averaging, we identified *Tetrahymena* RSP orthologs and solved the composition of individual RSs, showing their subunit heterogeneity. We found that RSP3 proteins are components of RS1 and RS2 but not RS3. Based on the presence of the RSP3 paralog, we distinguished sub-types of RS1 (RSP3A- or RSP3B-containing) and RS2 spokes (RSP3B- or RSP3C-containing). We identified novel RS-associated proteins, including several enzymes that may locally regulate ADP/ATP levels, GMP-recycling-related enzymes, and enzymes regulating phosphorylation levels. These discoveries will help to better understand the molecular mechanism(s) that regulate cilia beating and overall cilia metabolism.

Impact Statement

Identification of the subtypes of RS1 and RS2 spokes and RS1-3-specific RSPs. Discovery of the novel radial spoke structural components and RS-associated enzymes regulating ADP/ATP ratio and protein phosphorylation.

eLife assessment

This study presents **valuable** new insights from the protist *Tetrahymena* regarding radial spokes, conserved protein complexes that are **important** for cilia motility. The work employs interdisciplinary approaches to provide **convincing** support for radial spoke composition with some experiments, but there are weaker areas with partially **incomplete** support, such as relying on knockouts alone rather than including localization studies of tagged proteins.

Introduction

Motile cilia and homologous flagella are highly evolutionarily conserved, hair-like cell protrusions, supported by a microtubular skeleton, the axoneme, composed of nine peripheral doublets and two central singlets ($9 \times 2 + 2$). The central microtubules, C1 and C2, together with attached complexes, the projections, form the central apparatus (CA), the structure believed to initiate signals required for planar cilia beating. Transient interactions between CA projections and outer doublet-docked radial spokes (RSs) enable transmission of regulatory signals to outer doublet complexes, including the nexin-dynein regulatory complex (N-DRC) and the motor protein-containing outer and inner dynein arms (ODAs and IDAs) (Grossman-Haham et al., 2021 [DOI](#); Oda et al., 2014 [DOI](#); Smith & Yang, 2004 [DOI](#)).

Radial spokes are T-shaped, approximately 40-nm long complexes with three morphologically distinct regions: an elongated elastic stalk docked to the axoneme, an orthogonal head, and a neck connecting the stalk and the head. Within the axonemal repeat unit, radial spokes are arranged as triplets (RS1, RS2, RS3) within which the distance between the neighboring spokes is specific to each spoke pair, enabling their identification (Goodenough & Heuser, 1985 [DOI](#)). The architectures of RS1, RS2, and RS3 are not identical (Barber et al., 2012 [DOI](#); Lin et al., 2012 [DOI](#), 2014 [DOI](#); Pigino et al., 2011 [DOI](#); Yoke et al., 2020 [DOI](#)). The most striking example of the RS morphological diversity is a *Chlamydomonas* short, knob-like RS3, unfit to directly interact with the CA (Lin et al., 2012 [DOI](#); Pigino et al., 2011 [DOI](#)).

Pioneering studies using the *Chlamydomonas pf14* mutant carrying a mutation in the *RSP3* gene (Williams et al., 1989 [DOI](#)) and lacking full-length spokes, RS1 and RS2, led to the identification of radial spoke subunits, RSP1-RSP23 (Yang et al., 2001 [DOI](#), 2006 [DOI](#)). The list of *Chlamydomonas* RSPs was recently extended and most of the RSPs were assigned to RS1 and/or RS2 spokes (Gui et al., 2021 [DOI](#)). The RS1 protein composition was also resolved for the murine sperm axoneme (Zhang et al., 2022 [DOI](#)). However, despite the extensive research conducted using diverse ciliated species, including *Chlamydomonas*, *Tetrahymena* (Urbanska et al., 2015 [DOI](#); Vasudevan et al., 2015 [DOI](#)), *Ciona intestinalis* (Padma et al., 2003 [DOI](#); Satouh et al., 2005 [DOI](#); Satouh & Inaba, 2009 [DOI](#)), and mice (Guan et al., 2010 [DOI](#); Zhang et al., 2021 [DOI](#), 2022 [DOI](#)), the RS protein composition is still not fully solved, especially that of RS3. The knob-like RS3 is present in the *pf14* *Chlamydomonas* mutant (Lin et al., 2012 [DOI](#); Pigino et al., 2011 [DOI](#)) and thus, its protein composition was not resolved in earlier studies. In organisms where RS3 is a full-length spoke, its architecture differs from that of RS1 and RS2 (Chen et al., 2023 [DOI](#); Lin et al., 2012 [DOI](#); Pigino et al., 2011 [DOI](#)). To date, CFAP61 and CFAP251 are the only known RS3 components, building a part of the stalk and an arch-like structure at the RS3 base, respectively (Urbanska et al., 2015 [DOI](#)). This raises a question about the protein composition of this spoke.

The identification of RSPs and RSP-interacting proteins is crucial to understand the mechanism(s) of signal transduction from the CA to dynein arms. Here, we used the existing *Tetrahymena* knockout mutants with RS(s) defects (CFAP61-KO (Urbanska et al., 2015 [DOI](#)), CFAP206-KO (Vasudevan et al., 2015 [DOI](#)), and CFAP91-KO) (Bicka et al., 2022 [DOI](#)), and newly engineered mutants lacking RSP3 paralogs to decipher the protein composition of each RS in the triplet by label-free and tandem mass tag (TMT) quantitative mass spectrometry of wild-type and mutant ciliomes. Next, using co-immunoprecipitations and BioID, we verified that the identified proteins are either the subunits of RS complexes or are present in their vicinity. These proteomic approaches combined with cryo-electron tomography (cryo-ET) and subtomogram averaging studies of the ultrastructural alterations in the RS mutants allowed us to assign with high probability the substantial number of known RSPs and newly identified candidate proteins to either RS1, RS2, or RS3. Importantly, among the newly identified proteins, we found not only structural but also enzymatic proteins, including serine-threonine kinases, GMP recycling-related enzymes, adenylate kinases that dock to specific RSs, and an arginine kinase that locally controls the ATP level. We also noted a destabilization of some single-headed inner dynein arms in the analyzed RSP mutants.

As a result, we propose a model of the protein composition of *Tetrahymena* RSs showing both ubiquitous and RS-specific RSPs, new candidates of RS structural proteins, and RS-associated enzymes. The presence of spoke-specific RSPs agrees with structural differences among the individual RSs detected using cryo-ET (Lin et al., 2012 [DOI](#); Pigino et al., 2011 [DOI](#)). Moreover, we found that in *Tetrahymena*, there are sub-types of RS1 and RS2 spokes, which further increases the heterogeneity of radial spokes. Together, obtained data contribute to a better understanding of the molecular mechanisms regulating planar cilia motion.

Results

Deletion of different RSP3 paralogs causes heterogeneous phenotypic and ultrastructural defects

The RSP3 dimer is the main component of RSs (Wirschell et al., 2008 [DOI](#)). It stretches from the RS base (N-terminal end) to the RS head (C-terminal end) and directly interacts with numerous RSPs (Gui et al., 2021 [DOI](#); Oda et al., 2014 [DOI](#)). Comparative analyses of flagellar proteins purified from the wild-type and *pf14*, an RS1-RS2 spokes-less *Chlamydomonas* mutant (Yang et al., 2006 [DOI](#)) carrying a mutation in the *RSP3* gene (Williams et al., 1989 [DOI](#)) and cryo-EM studies combined with proteomics, and molecular modeling (Gui et al., 2021 [DOI](#)) yielded a model of RS1- RS2 protein composition in this green alga. We used a similar approach to decipher the protein composition of the *Tetrahymena* RSs. In contrast to *Chlamydomonas*, which has a single RSP3 gene, the *Tetrahymena* genome encodes three RSP3 proteins: RSP3A, RSP3B, and RSP3C (Figure 1A [DOI](#)). RSP3C is unusual in having an N-terminal ARF (ADP-ribosylation factor) domain and a C-terminal region enriched in glutamic acids. Such unusual RSP3 orthologs were also found in the proteomes of *Paramecium*, *Ichthyophthirius*, and *Pseudocohnilembus* belonging to Oligohymenophorea.

Tetrahymena RSP3 proteins expressed as C-terminal 3xHA-tagged fusions under the control of native transcriptional promoters localize uniformly along the entire cilia length (Figure 1B-E [DOI](#)). To explore whether the distribution of the three RSP3 paralogs differs between RS1, RS2, and RS3, we constructed knockout strains lacking single paralogs (Figure 2-figure supplement 1 [DOI](#)) using a germ-line approach (Cassidy-Hanley et al., 1997 [DOI](#); Dave et al., 2009 [DOI](#)). Compared to wild-type cells, RSP3 knockout cells swam significantly more slowly (Figure 2A-E [DOI](#)). The loss of RSP3B had the most profound effect and reduced the swimming rate to ~12% of the wild-type rate while deletions of either *RSP3A* or *RSP3C* were less damaging (70% and 56% of the wild-type rate, respectively).

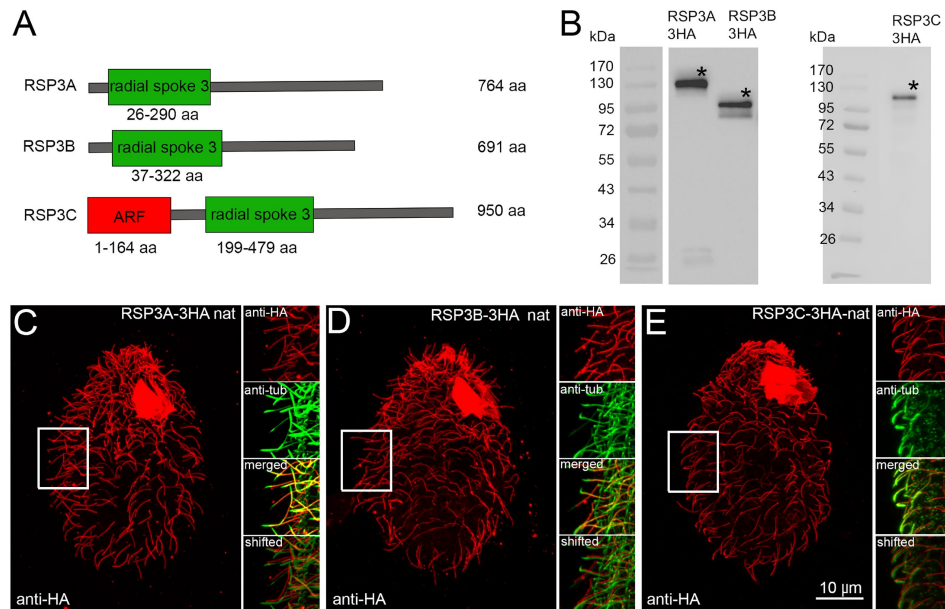


Figure 1.

Localization of RSP3 fusion proteins. (A) Domain organization of RSP3 paralogs predicted by SMART (<http://smart.embl-heidelberg.de/>). (B) Western blot analyses of ciliary proteins isolated from cells expressing RSP3 paralogs as C-terminal 3-HA tagged proteins under the control of native promoters. A star indicates a band corresponding to the position of fusion proteins, RSP3A-3HA, RSP3B-3HA, and RSP3C-3HA. (C-E) Ciliary localization of the RSP3 paralogs expressed as C-terminally -3HA tagged fusions under the control of the respective native promoters localize along the entire cilia length except for the ciliary tip. (C) RSP3A-3HA, (D) RSP3B-3HA, and (D) RSP3C-3HA labeled with a mix of anti-HA and anti-acetylated tubulin antibodies to visualize the entire cilia. To the right, the magnified fragments of the cells as indicated by the white insets, stained with anti-HA (red), anti-acetylated α -tubulin (green), and merged images (red and green). Below are images showing both channels but with some shifts to better visualize the presence of the RSP3 paralogs along the entire cilia. Note that all RSP3 paralogs localize along the entire cilia length except for a ciliary tip.

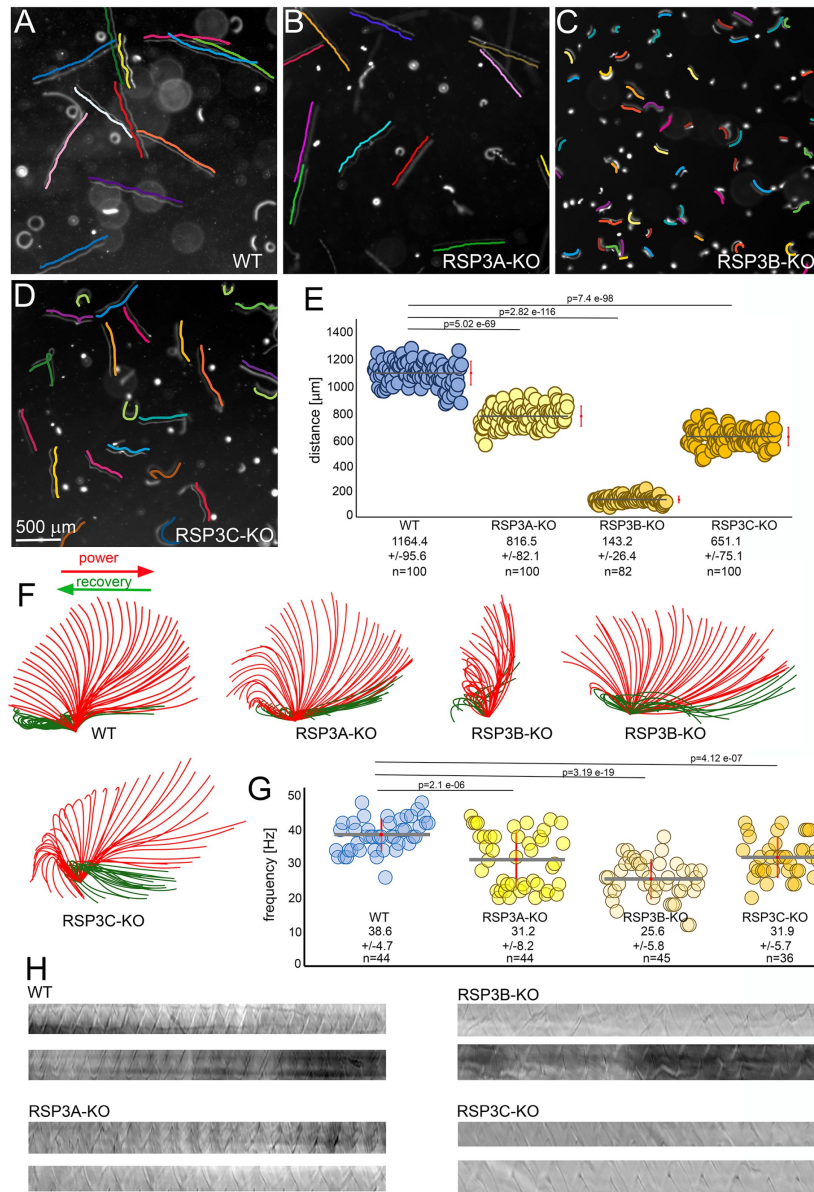


Figure 2.

Knockout of particular RSP3 paralogs affects cilia beating to different extents. (A-D) WT (A), RSP3A-KO (B), RSP3B-KO (C), and RSP3C-KO (D) trajectories recorded for 3 sec using a high-speed video camera. The cell swimming paths are indicated by the parallel colored lines. Bar = 500 μm. (E) A comparison of the length of trajectories recorded for 3 sec. The red bar position to the right represents the standard deviation. The wild-type cells swam on average 388 ± 32 μm/s ($n = 100$), and the swimming speeds of the RSP3A-KO, RSP3B-KO, and RSP3C-KO mutants were reduced to 272 ± 27 μm/s ($n = 100$), 48 ± 9 μm/s ($n = 82$), and 217 ± 25 μm/s ($n = 101$), respectively. Statistical significance was calculated using student t-test. (F) Drawings showing examples of the most frequently observed subsequent positions of a cilium of WT and RSP3 knockout cells. The ciliary waveform and amplitude in the RSP3B-KO mutant can vary both in the case of neighboring cilia and during subsequent cycles of the same cilium. (G) Graph showing a range of cilia beating frequencies in wild-type and RSP3 knockout mutants. The red bar represents the standard deviation. Statistical significance was calculated using student t-test. (H) Examples of the kymographs used to calculate cilia beating frequency.

Tetrahymena cells are propelled by the forces generated by cilia beating as metachronal waves. The cilium beating cycle has two phases, the power and recovery strokes that occur in different planes in relation to the cell surface, perpendicular and parallel, respectively. During the power stroke the cilium is straight while during the recovery stroke it bends, and the bend position shifts from the cilium base to the tip as the recovery stroke progresses (Figure 2F) (Bazan et al., 2021; Bicka et al., 2022). Deletion of *RSP3A* did not apparently affect the ciliary waveform, beat amplitude (Figure 2F), and cilia metachrony (Figure 2-figure supplement 2). In contrast, cilia lacking *RSP3B* were beating in a less coordinated manner, with inconsistencies in the waveform and amplitude between neighboring cilia and between subsequent beating cycles performed by the same cilium (Figure 2F). Moreover, in the majority of *RSP3B*-KO cilia their distal part remained bent during the power stroke (Figure 2F). A delayed straightening of the distal end during the power stroke was also observed in cells lacking *RSP3C*. However, in contrast to *RSP3B*-KO mutant, in *RSP3C* knockout, cilia beat with metachrony and beating amplitude was only slightly reduced (Figure 2F and Figure 2-figure supplement 2).

Compared to cilia in wild-type cells, which beat with a frequency of 38.6 ± 4.7 Hz, in *RSP3A* and *RSP3C* knockouts, the ciliary beating frequency was mildly reduced (on average 31.2 ± 8.2 Hz and 31.9 ± 5.7 Hz, respectively). In contrast, the loss of *RSP3B* reduced ciliary beating frequency nearly by half (25.6 ± 5.8 Hz) (Figure 2G-H).

The differences in the *RSP3* mutant phenotypes suggest that lack of the particular *RSP3* paralogs likely results in different ultrastructural alterations. Indeed, cryo-ET and subtomogram averaging analyses of *RSP* knockout cilia revealed paralog-specific ultrastructural defects (Figure 3, Figure 3-figure supplement 1, and Figure 3-figure supplement 2). The *RSP3A*-KO mutant cilia frequently (~35% of the 96-nm axonemal units) lacked either the entire RS1 (~30%, $n=639$ units) or RS1 except for the RS1 base (~67%, $n=1417$) (Figure 3 and Figure 3-figure supplement 1). The *RSP3B*-KO cilia lacked the entire RS2 structure in all analyzed 96-nm axonemal repeats. Moreover, 77.5% of the axonemal repeats ($n=1622$) showed RS1 spoke defects and lacked either the entire RS1 (26%) or, more frequently, its distal fragment (51.5%) (Figure 3 and Figure 3-figure supplement 1). Surprisingly, RS2 was also impaired in the majority of *RSP3C*-KO axonemal repeats, where it was either missing (20%) or only the RS2 base was well visible (62%). Thus, the RS2 head visible in the average image (Figure 3A) is due to the presence of intact RS2 spokes in 18% of the analyzed axonemal repeats (Figure 3 and Figure 3-figure supplement 1). Interestingly, in all *RSP3* mutants, the RS3 structure was intact, strongly suggesting that neither of the *RSP3* paralogs is a *Tetrahymena* RS3 component.

Taken together, the cryo-ET data suggest that in *Tetrahymena* there are at least two types of RS1 spokes: (i) RS1 type I containing *RSP3A* (spokes missing in *RSP3A*-KO cilia), making up approximately one-third of ciliary RS1 and (ii) RS1 type II, supported by *RSP3B* (missing in *RSP3B*-KO cilia), constituting approximately two-thirds of cilia RS1, and at least two types of RS2: (i) RS2 type I containing only *RSP3B* (3B-3B homodimer, intact RS2 spokes (18%) in *RSP3C*-KO cilia) and (ii) RS2 type II, having *RSP3C* in the stalk, likely as a heterodimer with *RSP3B*. However, we cannot exclude that some RS1 spokes can contain the *RSP3A*/*RSP3B* heterodimer or that *RSP3A* and *RSP3B* could partly substitute one another in the RS1 structure. Moreover, we cannot rule out that there is a minor fraction of RS2 containing the *RSP3C* homodimer. We must also keep in mind that in *Tetrahymena* the radial spokes' heads are connected and therefore the radial spokes may stabilize each other. Thus, some RS1 spokes could be destabilized in the *RSP3B*-KO mutant (secondary effect), as a moderate reduction in the level of *RSP3A* shown in global proteome comparisons (see below) coincided with the *RSP3B* deletion.

Identification of *Tetrahymena* radial spoke subunits by comparative mass spectrometry analyses of wild-type and RSP3 knockout ciliomes

To identify RS1, RS2, and RS3-specific components, we isolated cilia from wild-type cells, *RSP3* knockouts, and previously described *Tetrahymena* RSP knockouts: CFAP206-KO, missing either the entire RS2 or only its base, rarely accompanied by RS3 defects (Vasudevan et al., 2015 [\[1\]](#)), CFAP61-KO, lacking a part of the RS3 stalk (Urbanska et al., 2015 [\[2\]](#)), and CFAP91-KO frequently missing the RS3 and a part of the RS2 base (Bicka et al., 2022 [\[3\]](#)), and performed comparative proteomic mass spectrometry analyses using label-free and TMT 10-plex isobaric mass tagging approaches. The experiments were repeated three times to generate independent sets of data (at least 75% of proteins were present in all three replicates). Principal component analysis of the radial spokes mutant ciliomes using Ingenuity Pathway Analysis showed that the RSP3A-KO ciliome is most similar to the wild-type ciliome (**Figure 4A** [\[4\]](#)). The changes in the protein levels were analyzed using Perseus software (v. 2.0.3) (Tyanova et al., 2016 [\[5\]](#)).

Approximately one-thousand ciliary proteins from wild-type cells and RSP mutants were analyzed in LFQ experiments, while nearly four times more proteins were detected using the TMT approach. To determine the significant differences in the abundance of ciliary proteins between wild-type and knockout samples, we performed a t-test ($p < 0.05$) (**Figure 4** [\[4\]](#)-Source Data 1', **Figure 4** [\[4\]](#)-Source Data 2') and identified a number of proteins differentially abundant (with a ± 1.5 -fold change) in knock-out mutants compared to the wild-type control. The obtained data are graphically represented as Volcano plots $-\log(P \text{ value})$ vs. $\log_2$ (fold change of mutant/WT, **Figure 4A** [\[4\]](#)) and Venn diagrams (**Figure 5** [\[6\]](#)). The analyses of the levels of the outer dynein arms, two-headed, inner dynein arm IDA1/f, tether/tetherhead complex, nexin-dynein regulatory complex, and central apparatus components showed that these structures were generally unaffected (**Figure 4** [\[4\]](#)-Source Data 1', **Figure 4** [\[4\]](#)-Source Data 2').

To support and validate the global proteome comparisons, we performed co-IP and BioID assays followed by mass spectrometry analyses to identify ciliary proteins that either directly or indirectly interact with well-evolutionarily conserved, HA-tagged RSPs (**Figure 6** [\[7\]](#)-Source Data 1'), or are biotinylated in cilia of cells expressing RSPs fused with mutated biotin ligase, BirA* (**Figure 6-figure supplement 1** [\[8\]](#), **Figure 6** [\[7\]](#)-Source Data 2'). Next, we collated all proteomic data with bioinformatics analyses and cryo-ET ultrastructural studies to assign known RSPs and newly identified proteins to RS1, RS2, and RS3 spokes (**Figure 6** [\[7\]](#), **Figure 6-figure supplement 3** [\[9\]](#), and **Table 1** [\[10\]](#)).

Localization of the RSP3 paralogs

Changes in the RSPs levels in the analyzed mutants are presented in **Figure 4** [\[4\]](#)-Source Data 1' and **Figure 4** [\[4\]](#)-Source Data 2' and summarized in **Figure 6** [\[7\]](#), **Figure 6-figure supplement 3** [\[9\]](#), and **Table 1** [\[10\]](#). Global mass spectrometry analyses of the RSP3A-KO, RSP3B-KO, and RSP3C-KO axonemes confirmed that the protein encoded by the targeted gene was completely eliminated. Moreover, the knockout of *RSP3A* or *RSP3C* did not affect the levels of other RSP3 paralogs while in RSP3B-KO cilia the level of RSP3A was mildly reduced. The knockout of *CFAP61* has no effect on the level of RSP3 paralogs while in CFAP91-KO and CFAP206-KO mutants, the level of RSP3B and RSP3C were reduced, which agrees with the lack of RS2 spokes or part of the RS2 base/stalk in these mutants (Bicka et al., 2022 [\[3\]](#); Vasudevan et al., 2015 [\[2\]](#)). Thus, the comparative mass spectrometry analyses of wild-type and mutant ciliomes support our hypothesis that there are two types of RS1 spokes (with either RSP3A-RSP3A or RSP3B-RSP3B dimers). The reduced RSP3A level in RSP3B-KO mutant cilia suggests either that in the absence of RS2, some of the RSP3A-containing RS1 spokes are less stable or that some of RS1 spokes contain the RSP3A-RSP3B heterodimer.

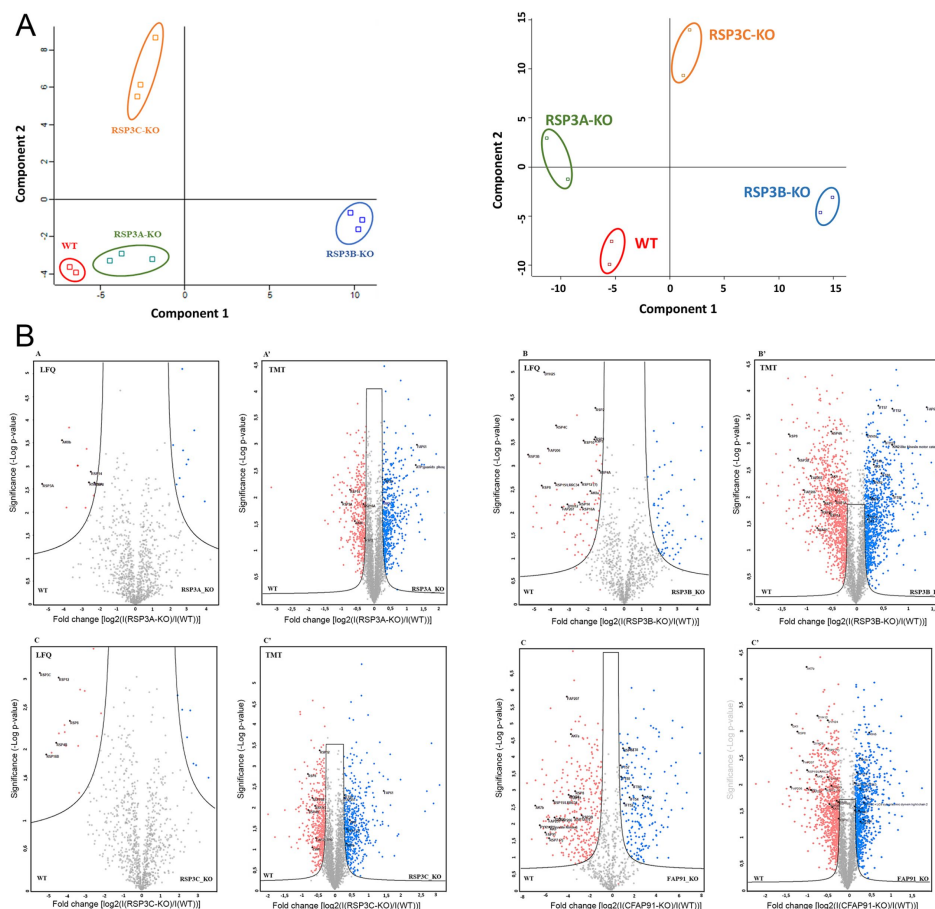


Figure 4.

Graphical illustration of quantitative proteomics data. (A) Principal component analysis (PCA) score plot from fold-change values of the differentially expressed proteins showing similar grouping of ciliomes identified by mass spectrometry using either LFQ (graphs to the left) or TMT (graphs to the right) strategy based on two independent variables. The analyzed ciliary protein samples were color-coded as follows: WT - red, RSP3A-KO - green, RSP3B-KO - blue, and RSP3C-KO - orange. (B) Volcano plots of wild-type and RS mutant ciliary proteins identified by mass spectrometry using either LFQ (graphs to the left) or TMT (graphs to the right) strategy (N=3) showing changes in mutant ciliomes. FDR=0.05. Proteins Red dots represent proteins that are at a significantly lower level in RSP mutants compared to wild-type cells, while blue dots represent proteins that are more abundant in mutant cilia. Names are provided for known RSP proteins. The statistical significance of the difference in protein enrichment between wild-type and RS mutant ciliomes calculated using Student's t-test.

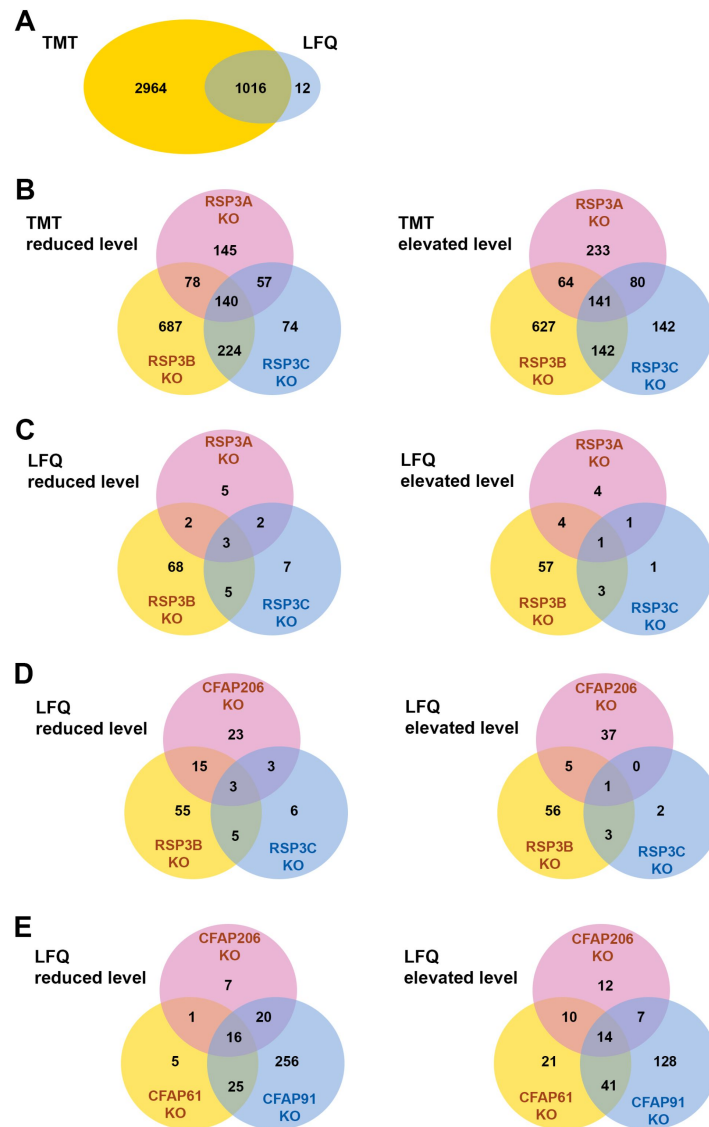


Figure 5.

Venn diagrams indicating the overlap of differentially expressed proteins between different radial spoke mutants as estimated by global ciliomes obtained either by LFQ or TMT approaches. (A) Comparison of the total number of proteins identified using LFQ and TMT approaches (B-E). The overlap of proteins that were either reduced (to the left) or elevated (to the right) in different radial spoke mutants as detected using (B) TMT and (C-E) LFQ methods. Images were prepared in Adobe Photoshop based on data sorted using the Excel program.

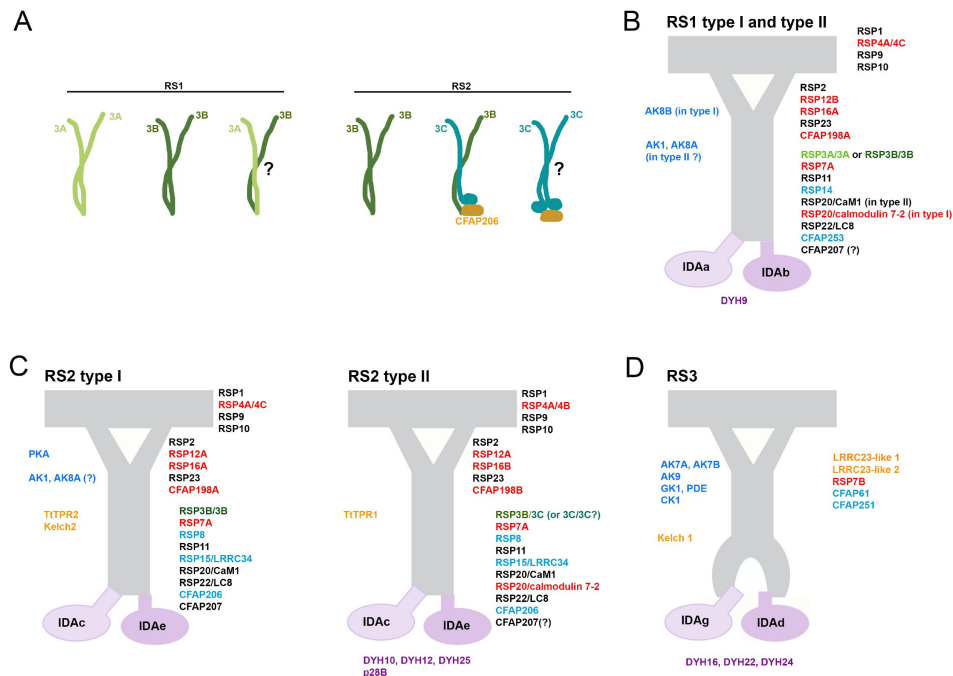


Figure 6.

A schematic summary of the radial spoke protein composition in *Tetrahymena* cilia prepared based on collected ultrastructural and proteomic data. (A) A summary of the RSP3C-RSP3C homodimer-containing RS2 and RSP3A-RSP3B heterodimer-containing RS1. RSP3A-RSP3B heterodimer-containing RS1 and RSP3C-RSP3C homodimer-containing RS2 are less probable. (B-D) Graphical presentation of the radial spokes composition, (B) RS1 containing either RSP3A or RSP3B homodimers, (C) RS2 containing either RSP3B homodimer or RSP3B-RSP3C heterodimer, (D) RS3. Names of the RS-specific RSPs existing in *Tetrahymena* as a single ortholog are in light blue; names of the RS- type specific RSP paralogs are in red; new candidate proteins are in orange; RS-associated enzymes are in blue. Note that in the case of RS2-type II (RSP3C-containing), RSPs were reduced in RSP3C-KO cilia and thus likely bind with RSP3C. DYH- dynein heavy chains.

RS1 3A-3A	based on	RS1 3B-3B	based on	RS2 3B-3B	based on	RS2 3B-3C	based on	RS3	based on
RS head									
RSP1	BioID: 3A, co-IP: 3A	RSP1	3B-KO ↓ BioID: 3B, 4A, 4C	RSP1	3B-KO ↓ BioID: 3B, 4A, 4C	RSP1	3B-KO ↓ BioID: 3B, 4A, 4C	LRRC23like 1?	91-KO ↓ but 3B-KO ↑ 3C-KO ↓ Hwang et al., 2023
RSP4A	BioID	RSP4A	3B-KO ↓ BioID	RSP4A	3B-KO ↓ BioID	RSP4A	BioID		
RSP4C	BioID	RSP4C	3B-KO ↓ BioID	RSP4C	3B-KO ↓ BioID	RSP4B	3C-KO ↓ BioID		
RSP9	BioID	RSP9	3B-KO ↓ BioID	RSP9	3B-KO ↓ BioID	RSP9	BioID		
RSP10	BioID	RSP10	3B-KO ↓ BioID	RSP10	3B-KO ↓ BioID	RSP10	BioID		
RS neck									
RSP2	3A-KO ↓ BioID: 3A, co-IP: 3A, 4A, 4C	RSP2	3B-KO ↓ BioID: 3B, 4A, 4C co-IP: 4A, 4C	RSP2	3B-KO ↓ BioID: 3B, 4A, 4C co-IP: 4A, 4C	RSP2	3B-KO ↓ BioID: 3B, 4A, 4C co-IP: 4A, 4C		
RSP12B	3A-KO ↓	RSP12B	3B-KO ↓	RSP12A	3B-KO ↓	RSP12A	3C-KO ↓		
RSP16A	3A-KO ↓ BioID	RSP16A	3B-KO ↓ BioID	RSP16A	3B-KO ↓ BioID	RSP16B	3C-KO ↓ BioID		
RSP23 CFAP67B CFAP198A	colP 3A-KO ↓ BioID	RSP23 CFAP67B CFAP198A	colP BioID 3B-KO ↓ BioID	RSP23 CFAP67B CFAP198A	colP BioID 3B-KO ↓ BioID	RSP23 CFAP67B CFAP198B	colP BioID 3C-KO ↓ BioID: CFAP206		
RS stalk and base									
RSP3A	3A-KO ↓	RSP3B	3B-KO ↓	RSP3B	3B-KO ↓	RSP3C	3C-KO ↓		
RSP7A	BioID: 3A	RSP7A	3B-KO ↓ BioID	RSP7A	3B-KO ↓ BioID	RSP7A	BioID:3C	RSP7B	91-KO ↓ BioID:CFAP61
RSP11	3A-KO ↓	RSP11	3B-KO ↓	RSP11	3B-KO ↓	RSP11	3C-KO ↓	CFAP61	91-KO ↓ Urbanska et al., 2015
RSP14	3A-KO ↓	RSP14	3B-KO ↓	RSP8	3B-KO ↓	RSP8	3C-KO ↓	CFAP251	91-KO ↓ Urbanska et al., 2015
RSP20 ? CaM1		RSP20 CaM1	3B-KO ↓ BioID:3B Gui et al., 2021	RSP20 CaM1	3B-KO ↓ BioID:3B	RSP20/CaM1	3C-KO ↓		
CaM 7-2	3A-KO ↓					CaM 7-2	3C-KO ↓		
RSP22/LC8	RS1 base in stable in 3A- KO coIP:3A BioID:3A	RSP22/LC8	3B-KO ↓	RSP22/LC8	3B-KO ↓	RSP22/LC8	RS2 base in stable in 3C-KO		
				CFAP206	3B-KO ↓ Vasudevan et al., 2015	CFAP206	Vasudevan et al., 2015		
CFAP207?	RS1 base in stable in 3A- KO	CFAP207	3B-KO ↓	CFAP207	3B-KO ↓ CFAP206- KO1	CFAP207? ?	RS2 base in stable in 3A-KO		
CFAP253	BioID:3A	CFAP253	BioID:3B						
New RS candidate proteins									
				TtTPR2	3B-KO ↓	TtTPR1	3C-KO ↓		
				Kelch motif- containing 2	3B-KO ↓			Kelch motif- containing 1	91-KO ↓
				LRR- containing	3B-KO ↓			LRR- containing LRRC23- like2	61-KO ↓ 91-KO ↓ 61-KO ↓ 91-KO ↓
RS associated enzymes									
AK8B	3A-KO ↓	AK1	3B-KO ↓	AK1?	3B-KO ↓ (not in CFAP206- KO)			AK7A	91-KO ↓
		AK8A		AK8A?	3B-KO ↓ (not in CFAP206- KO)			AK7B	91-KO ↓
				NDK	3B-KO ↓	NDK	3C-KO ↓	AK9	91-KO ↓
				PKA	3B-KO ↓	PKA	3C-KO ↓	GK1	91-KO ↓
				Ca/CaM- dependent kinase	3B-KO ↓	Ca/CaM- dependent kinase	3C-KO ↓	Casein kinase	61-KO ↓ 91-KO ↓ 91-KO ↓ BioID

Table 1

RSP proteins assigned to particular radial spoke types based on bioinformatics, proteomic, and cryo-EM data.

Based on the cryo-EM studies, we also hypothesized that the RS2 spokes contain either the RSP3B-RSP3B homodimer or the RSP3B-RSP3C heterodimer (not excluding, however, the possibility of the existence of the RSP3C-RSP3C homodimer). The comparative proteomic analyses revealed that the level of RSP3B did not change in RSP3C-KO mutant cilia, suggesting that in the absence of RSP3C, the RSP3B from the RSP3B-RSP3C heterodimer is still attached to the axoneme without maintaining the T-shape RS2 structure (**Figure 3-figure supplement 1**). However, the unaffected level of RSP3C (and RSP3C-associated RSPs, see **Figure 6-figure supplement 3**) in RSP3B-KO mutant cilia missing all RS2 spokes is surprising. To sum up, we found that RSP3B is a dominant *Tetrahymena* RSP3 paralog, roughly present in two-thirds of the RS1 spokes and likely in all RS2 spokes.

The RSP3 dimer interacts with other RSPs, forming together the RS head, neck, and stalk (Gui et al., 2021; Pigino et al., 2011; Yang et al., 2006). The bioinformatics analyses revealed that the *Tetrahymena* genome encodes more than one ortholog of several RSPs (**Table 1** and **Figure 6-figure supplement 3**). Combined comparative ciliomes analyses, co-IP, and BioID assays allowed us to assign the RSPs to individual RSs (**Figure 6** and **Tables 1** and **2**). Of note, the level of proteins that are the subunits of both RS1 and RS2 spokes in the RSP3A-KO mutant can be below the significance threshold (a ± 1.5 -fold change), as only one-third of RS1 contains the RSP3A paralog. Such proteins were assigned to RS1 based on the BioID data.

The components of the *Tetrahymena* radial spoke stalk

RSP8 and RSP14 are ARM-like (armadillo-like) motif-containing proteins. The level of RSP8 was significantly reduced in mutants with RS2 defects (RSP3B-KO, RSP3C-KO, and CFAP206-KO) but not in RSP3A-KO cilia, which lack a substantial portion of RS1 spokes. In contrast, the level of RSP14 was reduced in mutants with RS1 defects, RSP3A-KO (RS1 type I) and RSP3B-KO (RS1 type II), but was unchanged in RSP3C-KO or CFAP206-KO mutants with only RS2 defects. The reduced level of RSP8 in mutants lacking RS2 and RSP14 in mutants missing RS1 agrees with the BioID assays showing the biotinylation of RSP8 in cells expressing RSP3B, RSP3C, or CFAP206 BirA* fusions and biotinylation of RSP14 in cells expressing either RSP3A-HA-BirA* or RSP3B-HA-BirA*. Thus, RSP14 and RSP8 are RS1 and RS2 proteins, respectively, which agrees with recent studies in *Chlamydomonas* (Gui et al., 2021) and murine sperm axonemes (Zhang et al., 2022).

RSP7 and RSP11 form a heterodimer near RSP14 or RSP8 (Gui et al., 2021). The *Chlamydomonas* RSP7, in addition to the RIIa domain (dimerization-anchoring domain of cAMP-dependent PK regulatory subunit) at its N-terminus, also has several EF hand motifs in the middle and C-terminal regions. A search of the *Tetrahymena* proteome using *Chlamydomonas* RSP7 as a query revealed two proteins with a weak similarity to *Chlamydomonas* RSP7, RSP7A, the 39 kDa protein containing N-terminal RIIa and C-terminal calmodulin-binding motif (IQ, isoleucine, glutamine) and 66 kDa RSP7B with an AKA28 (A-kinase anchoring protein) domain and two EF hands. The level of RSP7A was significantly reduced only in RSP3B-KO cells, but RSP7A was biotinylated in cells expressing either of the RSP3 paralogs as BirA* fusions. In contrast, the level of RSP7B was reduced in cilia lacking CFAP91. Accordingly, RSP7B was poorly biotinylated in cells expressing RSP3 paralogs as BirA* fusions but detected by a higher number of peptides among proteins biotinylated in CFAP61-HA-BirA* cilia, leading to the conclusion that this is likely an RS3-associated protein. In *Tetrahymena*, RSP11 is a small (7.9 kDa) basic protein (pI = 9) with a predicted RIIa domain and limited similarity to *Chlamydomonas* RSP11. The level of RSP11 was reduced in all RSP3 mutants.

RSP15/LRRC34 (leucine-reach repeat-containing) protein was previously described as an RS2-specific protein in *Chlamydomonas* (Gui et al., 2021). Indeed, the *Tetrahymena* RSP15 ortholog was nearly completely eliminated or significantly reduced in mutants with RS2 defects, in agreement with its RS2 localization.

Protein name	Numer in TGD	RSP3A-KO/WT				RSP3B-KO/WT				RSP3C-KO/WT			
		LFQ		TMT		LFQ		TMT		LFQ		TMT	
		q-value	difference	q-value	difference	q-value	difference	q-value	difference	q-value	difference	q-value	difference
DYH6	TTHERM 00688470	0.427619	-0.63	0.97248	0.00	0.829776	-0.11	0.0333644	0.19	0.42251	-0.58	0.22074	0.14
DYH7	TTHERM 00912290	0.63228	-0.26	0.92333	0.01	0.889118	0.06	0.0436144	0.17	0.552849	-0.36	0.25214	0.14
DYH8	TTHERM 00531870	nd		0.35423	-0.09	nd		0.794792	0.02	nd		0.05891	0.27
DYH9	TTHERM 00947430	0.212183	-2.35	0.01518	-0.47	0.19376	-1.15	0.138485	-0.11	0.202845	-2.54	0.13222	-0.19
DYH10	TTHERM 00420340	nd		0.24242	-0.12	nd		0.0211327	-0.26	nd		0.78056	-0.03
DYH11	TTHERM 00252430	0.278289	-1.14	0.64207	0.04	0.079717	-1.13	0.695276	-0.02	0.207409	-1.33	0.61464	-0.05
DYH12	TTHERM 00919540	0.391149	-0.81	0.77556	-0.03	0.016887	-3.02	0.0024172	-0.39	0.216267	-1.14	0.89791	-0.01
DYH14	TTHERM 00492830	0.627178	-0.26	0.86334	-0.02	0.81443	0.11	0.0682636	0.15	0.604414	-0.28	0.36991	0.10
DYH15	TTHERM 00433800	0.509668	-0.39	0.53573	0.06	0.98351	-0.01	0.035494	0.19	0.58223	-0.33	0.13251	0.19
DYH16	TTHERM 00558640	0.500578	-0.41	0.49759	0.06	0.663928	-0.19	0.0255092	0.21	0.478641	-0.44	0.13177	0.20
DYH17	TTHERM 00850620	nd		0.16265	-0.15	nd		0.322829	-0.06	nd		0.56919	0.06
DYH19	TTHERM 01027670	0.59496	-0.32	0.13983	0.16	0.650933	-0.20	0.0160624	0.26	0.456762	-0.57	0.1503	0.17
DYH20	TTHERM 00821980	nd		0.27734	-0.11	nd		0.546651	0.04	nd		0.04042	0.29
DYH22	TTHERM 00565600	0.739417	-0.17	0.89428	0.01	0.80461	-0.12	0.0433816	0.18	0.728791	-0.18	0.63557	0.05
DYH23	TTHERM 00355100	nd		0.568953	-0.05	nd		0.716902	0.02	nd		0.36156	0.12
DYH24	TTHERM 00193520	0.454189	-0.60	0.04083	0.26	0.0565	-1.13	0.261682	0.07	0.306765	-0.88	0.23358	0.15
DYH25	TTHERM 00774810	0.497702	-0.46	0.29773	0.11	2.36E-05	-4.25	0.004094	-0.77	0.227139	-1.28	0.19224	-0.16
p28A	TTHERM 00841210	0.62273	0.34	0.13922	0.17	0.140637	-1.09	0.574056	0.04	0.529654	-0.45	0.28815	0.11
p28B	TTHERM 00319990	0.918872	0.05	0.71511	0.04	0.186438	-0.70	0.0378964	-0.20	0.546171	-0.42	0.43061	0.11
p28C	TTHERM 01129720	0.931915	0.05	0.8939	0.013	0.978215	-0.016	0.0700761	0.16	0.478333	-0.44	0.59091	0.07

Table 2

Comparative analyses of the levels of inner dynein arms (IDAs) components in RSP3 mutants

In early radial spoke studies (Yang et al., 2001 [DOI](#), 2006 [DOI](#)), the name RSP20 was assigned to calmodulin and RSP22 to LC8 (cytoplasmic dynein light chain 2). In *Chlamydomonas* flagella, calmodulin binds to FAP253 between RS1 and RS1-docked IDAs (Gui et al., 2021 [DOI](#)). In *Tetrahymena*, the level of calmodulin RSP20/CaM1 was reduced in mutants with RS2 defects (RSP3B-KO and RSP3C-KO) and was biotinylated in cells expressing RSP3B-HA-BirA*, suggesting an association of CaM1 with RS2 and perhaps RSP3B-containing RS1. On the other hand, the RS1 base containing CFAP253 is preserved in RSP3A-KO mutants, and thus, we cannot exclude the possibility that CaM1 is also an RS1 type I protein. Interestingly, the so-called calmodulin 7-2 was moderately reduced in cells with *RSP3A* or *RSP3C* deletion, suggesting that this calmodulin-type protein is the RS1 type I (RSP3A-containing) and RS2 type II (RSP3C-containing) component.

In *Chlamydomonas* flagella, RSP22/LC8 is associated with RS1 (four homodimers) and RS2 (two homodimers) (Gui et al., 2021 [DOI](#)). *Chlamydomonas* RSP22/LC8 is most similar to *Tetrahymena* LC8, a protein reduced in RSP3B-KO cilia. Based on BioID assays, this protein is likely positioned in the vicinity of all RSP3 orthologs, CFAP206, and CFAP91, suggesting an association with both RS1 and RS2. It was recently proposed that RS stalks dock to the LC8 multimers (Gui et al., 2021 [DOI](#)). Thus, it is possible that some RSP22/LC8 multimers remain associated with the axoneme in *Tetrahymena* RSP3A and RSP3C mutants. Indeed, the subtomogram average of *Tetrahymena* RSP3A-KO and RSP3C-KO cilia revealed that the spoke base remained associated with tubule A (**Figure 3** [DOI](#) and **Figure 3-figure supplement 1** [DOI](#)), which explains why the level of LC8 was not reduced in these mutants.

Recent studies have led to the identification of additional RSPs. FAP207, a MORN motif-containing protein, localizes at the RS1 and RS2 base, and the IQ motif-containing FAP253, orthologous to *Ciona* CMUB116 (Zhu et al., 2017 [DOI](#)) and murine IQUB (Zhang et al., 2022 [DOI](#)), is an RS1 adaptor protein (Gui et al., 2021 [DOI](#); Zhang et al., 2022 [DOI](#)). *Tetrahymena* CFAP253 and *Chlamydomonas* FAP253 are smaller than their mammalian orthologs and lack the ubiquitin homolog domain predicted in IQUB. The level of *Tetrahymena* CFAP253 was unaltered in all studied mutants, but CFAP253 was biotinylated in cells expressing RSP3A or RSP3B fused with BirA*. Thus, CFAP253 is likely an RS1 adaptor also in *Tetrahymena* cilia. The *Tetrahymena* genome also encodes one ortholog of FAP207 (**Table 1** [DOI](#) and **Figure 6-figure supplement 3** [DOI](#)). The level of *Tetrahymena* CFAP207 was significantly reduced in cells with RS2 defects (except for RSP3C-KO cilia, where the RS2 base is present) but not in RSP3A-KO cilia (the RS1 base is also maintained in RSP3A-KO cilia).

The components of the *Tetrahymena* radial spoke head

The RS head consists of two identical lobes connected by an RSP16 dimer. In *Chlamydomonas*, each lobe is composed of RSP4 and RSP6 (paralogous proteins), RSP5, RSP9, the MORN motif-containing RSP1 and RSP10, a fragment of the neck protein, RSP2, and a C-terminus of RSP3 (Gui et al., 2021 [DOI](#); Pigino et al., 2011 [DOI](#); Yang et al., 2006 [DOI](#)). The *Tetrahymena* genome encodes three paralogs of RSP4/RSP6: RSP4A, RSP4B, and RSP4C, and single orthologs of RSP1, RSP9, and RSP10. The identification of RSP1 and RSP10 orthologs was possible only by a combination of bioinformatics and proteomic data, as the *Tetrahymena* genome encodes 129 proteins containing MORN motifs (Habicht et al., 2015 [DOI](#)). Only two of those proteins, named here RSP1 and RSP10, were identified among proteins that co-immunoprecipitated with HA-tagged RSP4 and were biotinylated in BioID assays. A blastp search with *Chlamydomonas* RSP1 or RSP10 as a query revealed that *Tetrahymena* RSP1 and RSP10 are more similar to *Chlamydomonas* RSP10 than RSP1.

The levels of RSP1, RSP9, and RSP10 were significantly reduced only in RSP3B-KO lacking all RS2 and two-thirds of RS1. The levels of several other MORN domain proteins, similar to RSP1/10, were also altered in the ciliomes of the analyzed mutants, but none of them was detected in either co-IP or BioID assays (**Figure 6-figure supplement 3** [DOI](#)). Thus, it is unlikely that these other MORN domain-containing proteins are RS head components.

The loss of RSP3B coincided with the reduced levels of RSP4A and RSP4C, while the level of RSP4B was reduced in cilia of *RSP3C* and *CFAP91* knockout mutants. Moreover, RSP4A and RSP4C were biotinylated in RSP3A-HA-BirA*- and RSP3B-HA-BirA*-expressing cells, while RSP4B was biotinylated in RSP4A-HA-BirA*-expressing cells. We propose that the head of RSs containing either RSP3A or RSP3B as a stalk component is composed of RSP:1, 4A, 4C, 9, and 10, while RS2 spokes containing RSP3C likely have RSP4B instead of RSP4C in the head. These conclusions must be considered with caution, as in *Tetrahymena*, the RS heads are connected (Lin et al., 2012 [↗](#); Pigino et al., 2011 [↗](#)), and thus, a subunit of one spoke expressed as a BirA* ligase fusion could biotinylate proteins not only within the same spoke head but also of the adjacent spoke.

A search of the predicted *Tetrahymena* proteome with *Chlamydomonas* RSP5 aldo-keto reductase as a query failed to identify an ortholog. Among proteins annotated as aldo-keto reductases in the Tetrahymena Genome Database (TGD), thirteen were found in our *Tetrahymena* ciliomes, and four were significantly reduced in the RSP3 knockouts. However, these proteins did not co-precipitate with RSPs and were not biotinylated in cilia of *Tetrahymena* cells expressing RSP-BirA* fusions (Table 1 [↗](#) and Figure 6-figure supplement 3 [↗](#)). The human genome also lacks RSP5. It is likely that RSP5 is a *Chlamydomonas* and related algae-specific protein.

The components of the Tetrahymena radial spoke neck

In *Chlamydomonas*, the V-shaped radial spoke neck is composed of RSP2, RSP12, RSP16, RSP23/FAP67, FAP198, and FAP385 (Gui et al., 2021 [↗](#)). *Tetrahymena* has single orthologs of RSP2 and RSP23, two paralogs of RSP16, (RSP16A and RSP16B), CFAP198, (CFAP198A and CFAP198B), and RSP12 (RSP12A and RSP12B), but lacks FAP385.

Tetrahymena and *Chlamydomonas* RSP2 and RSP23/FAP67B proteins are similar only within the N-terminal regions containing DPY-30 and NDK (nucleoside diphosphate kinase) domains, respectively. RSP2 is likely a component of RS1 and RS2, as its level was reduced in both RSP3A and RSP3B knockouts (Tables 1 [↗](#) and 2 [↗](#)). The level of RSP23/CFAP67B was not significantly altered in the analyzed mutants, but it was identified in co-IP and BioID assays. In addition to RSP23/CFAP67B, the *Tetrahymena* genome encodes CFAP67A (THERM_00266490), with a high similarity to *Chlamydomonas* RSP23. CFAP67A was detected neither in co-IP nor BioID assays. Based on recent single-particle cryo-EM and molecular modeling studies, FAP67 (likely CFAP67A) is an A-tubule MIP protein (Kubo et al., 2023 [↗](#)). Thus, we propose that RSP23/CFAP67B is, similar to in *Chlamydomonas*, an RS1 and RS2 component.

The level of RSP16A was significantly reduced in cells with deletion of *RSP3A* or *RSP3B*, while the RSP16B level was diminished in cells with deletion of *RSP3C*, *CFAP206*, and *CFAP91*. The presence of RSP16A and 16B in RS1 and RS2 is also supported by BioID data.

Chlamydomonas FAP198 docks near RSP14/RSP8 stalk proteins (Gui et al., 2021 [↗](#)). In *Tetrahymena*, the level of CFAP198A was significantly reduced in *RSP3A* and *RSP3B* knockouts, while CFAP198B was reduced in cells with *RSP3C* deletion, suggesting that CFAP198A is a component of RS1 and RSP3B-containing RS2, while CFAP198B is a subunit of RSP3C-containing RS2. Importantly, both CFAP198A and B were identified in co-IP (CFAP198A) and BioID assays (both paralogs).

The *Tetrahymena* genome encodes 15 proteins similar to RSP12, the peptidyl-prolyl cis- trans isomerase. Among the seven that were identified in the *Tetrahymena* ciliome, two are RSP12 candidates. THERM_01018330 (RSP12A) could be an RS2 component, as its level was reduced in cilia with RS2 defects (RSP3B-KO, RSP3C-KO, CFAP206-KO). RSP12A was also reduced in CFAP91-KO cells; however, it is not clear whether the detected reduction is due to RS2 elimination or RS3 defects. The level of the isomerase encoded by THERM_00466090 (putative RSP12B) was reduced in cells lacking RSP3A or RSP3B. Both proteins were detected in BioID assays by a very limited number of peptides.

New radial spoke candidate proteins

The global comparative proteomic analyses led to the identification of several proteins, the candidate RSPs or proteins positioned in the RS vicinity (**Table 1** [↗](#) and **Figure 6-figure supplement 3** [↗](#)). The levels of two tetratricopeptide repeat (TPR)-containing proteins, TtTPR1 and TtTPR2, were reduced in mutants lacking RS2, TtTPR1 in *RSP3C* knockout and TtTPR2 in *RSP3B* knockout. TtTPR1 was found among biotinylated proteins in the BioID assays, while TtTPR2 appeared once in a co-IP. Kelch repeat-containing proteins are involved in various biological processes (Gupta & Beggs, 2014 [↗](#)). The level of Kelch repeat-containing protein 1 (Kelch1) was diminished in the CFAP91-KO mutant. In agreement, the BioID and co-IP data also suggest that Kelch1 is an RS3 protein. Kelch2 was reduced in *RSP3B* and *CFAP206* knockouts, suggesting its association with RS2 spoke.

The levels of two leucine-rich repeat-containing (LRRC) proteins similar to LRRC23, LRRC23-like 1 and LRRC23-like 2, were reduced in CFAP91-KO cilia but unchanged in *RSP3* mutants, suggesting their association with RS3 spoke. The level of LRRC23-like 2 was also reduced in the CFAP61-KO mutant lacking only a fragment of the RS3 stalk but having a neck and a head. Thus, LRRC23-like 2 could be a component of the RS3 stalk. It was recently reported that LRRC23 is a subunit of the RS3 head in human and mouse sperm (Hwang et al., 2023 [↗](#)). Thus, LRRC23-like 1 is likely a conserved component of RS3. The level of another LRRC protein, TTHERM_01084360, was reduced in cilia lacking *RSP3B*, *CFAP206* or *CFAP91*, suggesting its presence in the RS2 vicinity. Interestingly, a protein with similarity to human LRRC45 (TTHERM_01220370) was found in both RSP precipitates and among biotinylated proteins, but its level in RSP mutant cilia was similar to that in the wild type, suggesting that LRRC45 is not a component of the RSs but is positioned in the RS vicinity.

We also found that the level of TTHERM_00557860 protein with the coiled-coil domain, orthologous to human CCDC104/FAP36/BARTL1, recently shown to be an Arl3-binding protein in the primary cilia (Lokaj et al., 2015), was also altered in RS mutants. The putative CCDC104 ortholog was poorly biotinylated in BioID assays, and its role or precise localization in cilia requires further analyses.

Radial spoke-associated proteins with enzymatic activity

ATP homeostasis

The ultrastructural defects in RS mutant cilia are accompanied by changes in the levels of a number of enzymes (**Table 1** [↗](#) and **Figure 6-figure supplement 3** [↗](#)), suggesting that RSs serve as enzyme docking sites. Dynein arm motor activity requires a constant supply of ATP and its uniform distribution within the entire cilium. ATP is both delivered to cilia from the cell body by diffusion and produced within cilia (Villar et al., 2017 [↗](#)). The levels of several enzymes involved in ATP homeostasis were altered in radial spoke mutants.

Enolase 1 (ENO1) and pyruvate kinase 1 (PYK1) are glycolytic enzymes that catalyze two final steps resulting in ATP synthesis. The levels of both enzymes were reduced in *RSP3B* and *CFAP61* knockouts (ENO1 also in CFAP91-KO). Intriguingly, the reduction of the ENO1 and PYK1 levels was also reported in the *Tetrahymena* central apparatus mutants, with the deletion of either *SPEF2A* or *CFAP69* genes encoding subunits of the C1b projection (Joachimciak et al., 2021 [↗](#)).

Adenylate kinases and phosphagen kinases can serve as ATP-shuttles and locally control ATP levels. Phosphagen kinases (creatine kinase in vertebrates, arginine kinase in *Tetrahymena*) catalyze the reversible transfer of the phosphate group from ATP to a guanidine substrate. The *Tetrahymena* genome encodes two arginine kinases, AK1 and AK2 (Michibata et al., 2014 [↗](#); Uda et al., 2006 [↗](#)). The levels of ~40 kDa AK1 and ~80 kDa AK2 were reduced in RS mutants except for *RSP3A*-KO. ENO1, PYP1, AK1, and AK2 were not identified in co-IP or BioID assays.

Adenylate kinases reversibly catalyze the $\text{ATP} + \text{AMP} \leftrightarrow \text{ADP}$ reaction and thus locally control the ATP level and maintain the homeostasis of adenine nucleotides (Dzeja & Terzic, 2009 [↗](#)). The levels of at least six adenylate kinases (types 1, 7, 8, and 9) were changed in the cilia of *Tetrahymena* RS mutants. Importantly, adenylate kinases were also found among proteins that either co-immunoprecipitated with RSPs and/or were biotinylated in cilia of cells expressing RSPs fused with mutated BirA* ligase. Thus, adenylate kinases are likely radial spoke-associated proteins, while changes in the levels of arginine kinase 2, ENO1 and PYK1 may reflect more general changes in cilia physiology.

The level of AK8B adenylate kinase was dramatically reduced in RSP3A-KO cilia, while kinases AK7B and AK9 (and to a lesser extent, AK7A) were nearly completely eliminated in cilia of the CFAP91-KO mutant but unaltered in RSP3 mutants. Thus, RS1 is likely the main docking site for AK8B, while AK7A, AK7B, and AK9 preferentially associate with the RS3 spoke or its vicinity. AK1 and AK8A are likely associated with RSP3B-containing spokes, as their levels were significantly reduced in cilia of RSP3B-KO but not RSP3A-KO or RSP3C-KO mutants. With the exception of AK7B, AK8B, and AK9, which were either completely missing or substantially reduced in the analyzed mutants, the levels of other adenylated kinases were only moderately reduced, suggesting that in addition to radial spokes, AK1, AK7A, and AK8A likely associate with other ciliary structures.

cGMP-related enzymes

Of the two *Tetrahymena* guanylate kinases, GK1 is present in the *Tetrahymena* ciliome. Guanylate kinase catalyzes the transfer of phosphate from ATP to GMP, producing GDP and ADP. The phosphorylation of GDP to GTP is catalyzed by nucleoside-diphosphate kinases (NDK), and GTP can be converted to cGMP by guanylate cyclase, while 3'5'-cyclic nucleotide phosphodiesterase (PDE) hydrolyzes cGMP (Muthaiyan Shanmugam et al., 2018 [↗](#); Wyatt, 2015 [↗](#); Wyatt et al., 1998 [↗](#)). The levels of GK1 and PDE (similar to AK9) were dramatically reduced in CFAP91-KO cilia (PDE also in CFAP61-KO) but not in RSP3 mutants. Moreover, GK1 co-immunoprecipitated with an RS3 base protein, CFAP251. Thus, GK1 and PDE are likely RS3-associated enzymes, and cGMP could play a role in signal transduction in the vicinity of RS3. The RSP23 protein contains NDK domains (Patel-King et al., 2004 [↗](#)); however, its level did not significantly change in the analyzed mutants. Interestingly, we identified another protein containing an NDK domain (TTHERM_000160999) whose level was reduced in cilia lacking RSP3B or RSP3C.

Early analyses of ciliary beating using another ciliate, *Paramecium*, as a model showed that cyclic nucleotide monophosphate-driven chemical signaling controls ciliary beating (Bonini & Nelson, 1988 [↗](#); Hemmersbach et al., 1998 [↗](#); Noguchi et al., 2004 [↗](#)). Thus, the immobility of the CFAP91-KO mutants might be caused not only by the lack of RS3 and some IDAs (Bicka et al., 2022 [↗](#)) but also by the perturbation of guanidine nucleotide homeostasis.

Serine/threonine protein kinases

Comparative global proteomic analyses also revealed that the levels of three serine/threonine kinases, casein kinase 1 (CK1), calcium/calmodulin-dependent protein kinase, and cAMP-dependent protein kinase (PKA), were altered in radial spoke mutants. In *Chlamydomonas*, CK1 docks to outer doublets and regulates the phosphorylation of IC138, the IDA1/f intermediate chain (Yang & Sale, 2000 [↗](#)). In *Tetrahymena*, the level of CK1 was reduced in CFAP91-KO and CFAP61-KO cilia, suggesting its association with RS3 spoke. The presence of CK1 in the RS3 vicinity was confirmed by co-IP and BioID studies; however, it is unlikely that RSs are the sole CK1 docking site.

The RSP3 protein has an AKAP domain (A-kinase anchoring domain) (Gaillard et al., 2001 [↗](#)) and thus locally regulates PKA localization and activity by restricting access to substrates. In *Chlamydomonas*, disruption of the RSP3 AKAP domain affects the activity of cAMP-dependent protein kinase (PKA) and causes abnormal flagellar motility (Gaillard et al., 2006 [↗](#)). PKA functions

as a tetramer composed of two regulatory and two catalytic subunits (Scott & Pawson, 2009 [↗](#)). The levels of two PKA catalytic subunits, THERM_00433420 and THERM_00658860, and a regulatory subunit (THERM_00623090) were reduced in RS2 mutants.

The level of calcium/calmodulin-dependent protein kinase was apparently diminished in mutants with RS2 defects and in CFAP91-KO cilia, suggesting a role of RS2 and perhaps RS3 in calcium/calmodulin-dependent signaling.

Interestingly, at least some RSPs are present in cilia in more than one isoform (2- dimensional analyses of the HA-tagged RSPs, **Figure 6-figure supplement 2** [↗](#) and) (Bicka et al., 2022 [↗](#)). It is tempting to speculate that some RSPs undergo phosphorylation and that the phosphorylation status controls other protein binding and/or signal transduction.

Radial spoke defect-related instability of inner dynein arms

Single-headed inner dynein arms (IDA) are docked in pairs at the radial spoke base, dynein a and b at RS1, c and e at RS2, and d and g at RS3. They are composed of IDA type- specific dynein heavy chain (DYH), actin, and centrin (dyneins b, e, g) or light chain protein, p28 (dyneins a, c, d) (Yamamoto et al., 2021 [↗](#)). The comparative global proteomic analyses revealed that the deletion of genes encoding the components of the RS stalk also affected the levels of some IDAs (**Table 2** [↗](#)). The level of DYH9 was reduced in RSP3A-KO cells, while the levels of DYH10, 12, and 25 were decreased in the RSP3B-KO mutant, which coincides with the lack of dynein c density in the 96-nm subtomogram averaged map of RSP3B-KO cilia. The levels of DYH10, 12, and 25 were also reduced in CFAP206-KO, specifically missing dynein c (Vasudevan et al., 2015 [↗](#)), and in CFAP91-KO mutants (Bicka et al., 2022 [↗](#)), further supporting the assumption that these dynein heavy chains are the components of dynein c. The most frequent ultrastructural defect in *Tetrahymena* CFAP91-KO mutant cilia is a lack of RS3. As we previously reported, in addition to DYH10, 12, and 25, the levels of DYH16, 22, and 24 are moderately reduced in CFAP91-KO cilia (Bicka et al., 2022 [↗](#)), suggesting that these dynein heavy chains are components of IDAd or IDAg.

Tetrahymena has three orthologs of the single-headed IDA light chain protein p28: p28A, p28B, and p28C (Subramanian et al., 2016 [↗](#)). The level of p28B was reduced in cells lacking RSP3B and CFAP206, which agrees with p28B being a dynein c subunit (Yamamoto et al., 2021 [↗](#)). In CFAP91-KO cilia, not only was the level of p28B reduced (Bicka et al., 2022 [↗](#)) but also that of p28A, suggesting that p28A is a main component of dynein d (Yamamoto et al., 2021 [↗](#)). The above data provide further experimental evidence regarding the protein composition of single-headed IDAs and show a heterogeneity of the protein composition of IDAs of a particular type (three different dynein heavy chains can be a component of IDAc). This raises a question about the biological significance of such diversity. Interestingly, in both RS2 mutants (RSP3B- KO and CFAP206-KO), only dynein c is missing (Vasudevan et al., 2015 [↗](#)) and this work), suggesting differences in the docking of dynein c and dynein e at the RS2 base.

Discussion

Radial spokes are important links in the signal transduction chain from the central apparatus to the dynein arms. Here, using bioinformatics, proteomic and cryo-ET approaches, we were able to define a large part of the protein composition of the individual RSs in the ciliate *Tetrahymena* (**Figure 6** [↗](#)). In contrast to other studied organisms assembling motile cilia or flagella, the genome of *Tetrahymena* encodes more than one ortholog of several RSP genes (RSP3, RSP4/6, RSP7, RSP12, RSP16, CFAP198). This finding agrees with the *Tetrahymena* genome analyses revealing duplication of some genes encoding proteins involved in sensing or structural complexity (Eisen et al., 2006 [↗](#)).

Immunofluorescence analyses revealed that *Tetrahymena* RSP3 and RSP4/6 paralogs are uniformly distributed along the entire cilium length. However, they are components of RSs of a specific type. Strikingly, in *Tetrahymena*, the differences in protein composition are not only between RS1, RS2, and RS3 spokes, as was shown earlier in *Chlamydomonas* (Gui et al., 2021 [DOI](#)) and mouse sperm axonemes (Zhang et al., 2022 [DOI](#)) but also within the group of RS1 and RS2 spokes. *Tetrahymena* likely has at least two types of RS1 (type I containing RSP3A and type II having RSP3B homodimers) and likely two types of RS2 spokes (type I containing RSP3B homodimer and type II with RSP3B-RSP3C heterodimer). At the moment, we cannot exclude the possibility that the RSP3A and RSP3B proteins also form heterodimers and that RSP3C can form homodimers. Interestingly, deletion of CFAP206 has a stronger impact on the level of RSP3C than RSP3B, while the level of CFAP206 is unaltered in RSP3C-KO but significantly reduced in RSP3B-KO, suggesting that RSP3C docks to CFAP206, which in turn binds to RSP3B. We have also found that some RSPs associate with a specific RSP3 paralog, further increasing RS diversity and suggesting co-evolution of some RSPs. Thus, the *Tetrahymena* cilium is a mosaic of RS1 and RS2 spokes composed of different RSP3 paralogs. At present, the significance of RS1 and RS2 heterogeneity is unknown.

The RS head transiently interacts with CA projections. These interactions can be mechano-chemical (Oda et al., 2014 [DOI](#)) and electrostatic in nature (Grossman-Haham et al., 2021 [DOI](#)). The very C-terminal end of RSP3C, compared to the RSP3A and RSP3B proteins, is enriched in glutamic acid residues that increase the RS head surface negative charge. If the RSP3C C-terminal end, similar to the *Chlamydomonas* RSP3 tail, is a part of the radial spoke head, its presence increases the negative charge of the head surface and thus might strengthen the electrostatic interactions with CA projections. Thus, the incorporation of RSP3C into the RS2 structure could locally strengthen the interactions between the RS and the CA projection(s). Interestingly, the N-terminal part of RSP3C is also unusual in having an ARF domain, which could play a role in signaling.

The analyses of the proteins biotinylated in cells expressing RSPs-BirA* fusions revealed the presence of several CA proteins (Figure 6 [DOI](#)-Source Data 2'), suggesting close proximity between the projections and the RSs. In particular, numerous peptides were those of two proteins, CFAP46 and CFAP54, the components of the C1d projection, which is surprising, as C1d is not the longest projection. This unexpected result calls for further studies.

The presented proteomic data clearly indicate that the lack of specific RSP3 protein causes not only destabilization of certain RSs but also affects the stability of specific IDAs, the receivers and effectors of transduced regulatory signals and perhaps other minor structures/linkers/proteins positioned in the RS vicinity. In cells with knocked out *RSP3B*, only one of two IDAs docked at the RS2 base, IDAc, is missing. These data strongly suggest that the docking of two IDAs, IDAc and IDAe, at the RS2 base is different. Moreover, comparative proteomic analyses of wild-type and RSP3B-KO cilia enabled the identification of dynein heavy chains that are components of IDAc.

The global comparative analyses of wild-type and RS mutant ciliomes led to another interesting discovery – the identification of several enzymes that likely dock to specific RSs. Among them are (i) adenylate kinases that together with arginine kinase locally regulate ATP homeostasis, (ii) enzymes involved in GMP recycling, and (iii) serine-threonine kinases that could play a role in the control/regulation of RS-mediated signal transduction. Thus, the phenotypic outcome of the deletion of genes encoding RS structural proteins is not simply a consequence of the radial spokes ultrastructural alterations but also of the alterations of the IDAs, linkers, and changes of the level of nucleotide, Ca²⁺/calmodulin, and protein phosphorylation-mediated signaling as well as the accessibility of the ATP-stored energy.

Cell growth conditions and phenotypic analyses

Tetrahymena cells were grown to the mid-log phase with shaking (80–110 rpm) at 30 °C. Wild- type cells and motile mutants were cultured in a standard SPP medium (Gorovsky et al., 1975 [DOI](#)), while mutants with major ciliary defects were grown in a rich MEPP medium (Orias & Rasmussen, 1976 [DOI](#)), both supplied with an antibiotic-antimycotic mix (Sigma-Aldrich, St. Louis, MO, USA) at 1:100 (SPP) or 1:50 (MEPP). The measurements of the cell swimming rate and the analyses of cilia beating (amplitude, waveform and frequency) were described in detail (Bicka et al., 2022 [DOI](#)). Briefly, for swimming rate analyses, cells at a density of 10^4 cells/mL were viewed and recorded at room temperature using a Zeiss Discovery V8 Stereo microscope (Zeiss, Oberkochen, Germany) equipped with a Zeiss Plans 10_ FWD 81 mm objective, and an Axiocam 506 camera and ZEN2 (blue edition) software. The length of the trajectories was measured using ImageJ software. To analyze cilia beating, cells at a density of 10^5 cells/mL were cultured at room temperature for 3 hours, and beating cilia were recorded using a Phantom Miro C110 high-speed camera (Vision Research, Wayne, NJ, USA) mounted on an AXIO Imager M2 microscope (Zeiss, Germany) with either a 40 x oil immersion lens (analyses of cilia beating frequency) or a 63x oil immersion lens (numerical aperture 1.4, analyses of ciliary waveform). Videos were recorded at 900 frames/s and analyzed using ImageJ as described (Bicka et al., 2022 [DOI](#)).

Knock-ins and knock-outs

Engineering and phenotypic analyses of *Tetrahymena* mutants with deleted *CFAP61* (Urbanska et al., 2015 [DOI](#)), *CFAP206* (Vasudevan et al., 2015 [DOI](#)), or *CFAP91* (Bicka et al., 2022 [DOI](#)) were described. The RSP3 genes' fragments used to obtain knock-out and knock-in transgenes were amplified by PCR with the addition of restriction sites using high-fidelity polymerase and the appropriate primers listed in Table S1. To delete fragments of the RSP3 genes, approximately 1.2–1.5 kb fragments positioned upstream and downstream of the targeted gene fragment were amplified and cloned subsequently on both sites of the neo4 resistance cassette (Mochizuki, 2008). Approximately 60 µg of plasmid was digested with *ApaI* and *SacII* to separate the transgene from the plasmid backbone, precipitated onto gold particles and used to transform conjugating *Tetrahymena* cells. Transformed cells were processed as described (Cassidy-Hanley et al., 1997 [DOI](#); Dave et al., 2009 [DOI](#)). The deletion of the targeted fragments was confirmed by PCR.

To express RSP proteins as fusions with C-terminal either -3HA or -HA-BirA* tag under the control of a respective gene promoter, approximately 1 kb fragments of the open reading frame without the stop codon and 1 kb fragment of the 3'UTR were amplified and cloned into CFAP44-3HA-neo4 or CFAP44-HA-BirA*-neo4 plasmids, respectively (Urbanska et al., 2018 [DOI](#)) to replace CFAP44 gene fragments. On average, 10–15 µg of plasmid was linearized with *MluI* and *XhoI*, precipitated onto gold, and used to transform CU428 cells.

Immunofluorescence

To localize HA-tagged RSPs, cells from the overnight culture were fixed on coverslips with a mix of 0.25% Triton-X-100 and 4% PFA in a PHEM buffer. After drying and blocking, cells were stained overnight with a mix of primary antibodies, monoclonal anti-HA antibody (BioLegend, San Diego, CA, USA) diluted 1:300 and 6-11 B1 (Sigma-Aldrich), an anti- acetylated α -tubulin antibody (1:2000) at 4 °C. After washing with PBS, samples were stained for 1.5 h at RT with a mix of the secondary antibodies, anti-mouse or anti-rabbit IgG, conjugated with either Alexa-488 or Alexa-555 (Invitrogen, Eugene, OR, USA) diluted 1:300. Coverslips were mounted in Fluoromount-G (Southern Biotech., Birmingham, AL, USA). Cells were recorded using either a Zeiss LSM780 (Carl Zeiss Jena, Germany) or a Leica TCS SP8 (Leica Microsystems, Wetzlar, Germany) confocal microscope.

Cryo-ET preparation

The axonemes for cryo-ET were cross-linked with glutaraldehyde (final concentration 0.15%) for 40 min on ice and quenched with 35 mM Tris pH 7.5. The axoneme solution at 3.6 mg/mL was mixed with 5 (Cytodiagnostics) or 10 (Aurion) nm gold beads in a 1:1 ratio for a final axoneme concentration of 1.8 mg/mL. Inside the Vitrobot Mk IV (Thermo Fisher) chamber, 4 μ l of crosslinked axoneme sample was applied to negatively glow discharged (10 mA, 10 s) C-Flat Holey thick carbon grids (Electron Microscopy Services #CFT312-100). The sample was incubated at 23 °C and 100% humidity for 45 seconds on the grid, followed by 8 seconds of blotting with force 0 and plunge frozen in liquid ethane.

Cryo-ET acquisition and reconstruction

Tilt series were collected using the dose-symmetric scheme from -60 to 60 degrees with an increment of 3 degrees at 2.12 Å per pixel using a Titan Krios equipped with Gatan K3 and BioQuantum energy filter. The acquisition was performed using SerialEM (Mastronarde, 2005 [\[1\]](#)). The defocus for each tilt series ranges from -2.5 to -6 μ m. The total dose for each tilt series is 120 to 160 e- per Å². For each view, a movie of 10-13 frames was collected. Motion correction of each view was performed with Alignframes (Mastronarde & Held, 2017 [\[2\]](#)). Tomograms were reconstructed using IMOD (Kremer et al., 1996 [\[3\]](#)).

Subtomogram Averaging

CTF estimation for each tilt series was performed with WARP (Tegunov & Cramer, 2019 [\[4\]](#)). The doublet microtubule for subtomogram averaging was picked using IMOD by tracing the line along the microtubules (Kremer et al., 1996 [\[3\]](#)). Subtomogram averaging of the 4-times binned 96 nm repeating unit of WT and mutant strains was performed using the “axoneme align” program (Bui & Ishikawa, 2013 [\[5\]](#)). The subtomogram coordinates and alignment parameters were converted to Relion 4.0 for local refinement and classification (Kimanius et al., 2021 [\[6\]](#)). The resolutions for the 96-nm repeating unit of the axoneme of wild type, *RSP3A-KO*, *RSP3B-KO*, and *RSP3C-KO* are 18, 20, 22, and 17 Å, respectively. Axonemal repeats from *RSP3A* and *RSP3B* mutants showed strong heterogeneity in the occurrence of RS1 and RS2. To analyze the heterogeneity of the mutant strains, three-dimensional (3D) classification without alignment was performed in Relion by masking RS2 and RS1. The axoneme of the *RSP3C* mutant strain showed heterogeneity in the RS2 head and neck region. Therefore, unsupervised 3D-classification by masking the head and neck region was performed with three classes.

For visualization, tomograms were CTF deconvolved and missing wedge corrected using IsoNet (Liu et al., 2022 [\[7\]](#)). The UCSF ChimeraX package was used for the visualization of subtomogram averages, surface rendering, segmentation, and fitting (Pettersen et al., 2021 [\[8\]](#)).

Deciliation and Western Blot

All buffers used in biochemical studies were supplied with protease inhibitors (cOmplete, EDTA-free protease inhibitor cocktail, Roche Diagnostics GmbH, Mannheim, Germany). To purify cilia, cells from the overnight, mid-log phase culture were collected, rinsed with Tris-HCl buffer, pH 7.4, resuspended in a deciliation buffer (10 mM Tris-HCl, pH 7.4, 10 mM CaCl₂, 50 mM sucrose), and deciliated by a pH shock (Wloga et al., 2008 [\[9\]](#)). Deciliation was monitored under a microscope. Samples containing intact non-moving cells were spun down twice at 1680 x g for 5 min to remove cell bodies, and cilia were collected by centrifugation at 23,300 x g for 30 min. Next, cilia were suspended in 10 mM Tris-HCl buffer (pH 7.4) with protease inhibitors, and the protein concentration was estimated using Pierce™ BCA Protein Assay Kit (Thermo Scientific, Bartlesville, OK, USA).

For Western blot analyses, 30 µg of ciliary proteins were loaded on the gel. After transfer to nitrocellulose, HA-tagged fusion proteins were detected using monoclonal anti-HA antibodies diluted 1:2000, followed by washing (4 x 10 min, TBST) with an incubation with HRP- conjugated secondary antibodies, 1: 10 000 in 5% milk/TBSP (Jackson ImmunoResearch, West Grove, PA, USA). HA-tagged proteins were detected using a Westar Supernova kit (Cyanagen, Italy). Two-dimensional electrophoresis was performed as described (Bicka et al., 2022 [DOI](#)).

Co-immunoprecipitation and proximity labeling (BioID) assays

For co-immunoprecipitation assay, 200 ml of cells (3×10^5 cells/ml) expressing -3HA tagged radial spoke proteins under the control of a native promoter and wild-type cells (control), all at mid-log phase, were spun down and washed with 10 mM Tris-HCl buffer (pH 7.4). After deciliation, collected cilia were suspended in 10 mM Tris-HCl buffer (pH 7.4) and combined with an equal volume of 2% NP-40 and 1.2 M NaCl in 80 mM Tris-HCl buffer, pH 7.5, with protease inhibitors. After a 15 min incubation on ice, axonemes were pelleted at $20,000 \times g$ and treated with 0.5 M KI, 30 mM NaCl, 5 mM MgSO₄, 0.5 M EDTA, 1 mM dithiothreitol, and 10 mM HEPES, pH 7.5. After 30 min on ice, the axonemes were centrifuged ($16,000 \times g$ for 15 min at 4 °C), and the collected supernatant was diluted with 50 mM Tris-HCl, pH 7.4 (1:9). The axonemal proteins were incubated overnight with anti-HA magnetic beads with bound anti-HA antibodies (Thermo Fisher Scientific, Waltham, MA) at 4 °C. The bead-bound proteins were analyzed by SDS-PAGE and silver staining and identified by mass spectrometry.

The proximity-labeling assay (Roux et al., 2012 [DOI](#)) was performed as described in detail (Joachimik et al., 2021 [DOI](#)). Briefly, 200 ml of cells (3×10^5 cells/ml) expressing RSP-HA-BirA* and wild-type cells (control) at mid-log phase were spun down and suspended in 200 mL of 10 mM Tris-HCl buffer (pH 7.4). After approximately 18 h, Tris buffer was supplied with biotin (50 µM final concentration), and the cells were cultured for 4 h at 30 °C. After deciliation, collected cilia were suspended in axoneme stabilization buffer (20 mM potassium acetate, 5 mM MgSO₄, 0.5 mM EDTA, 20 mM HEPES, pH 7.5 with protease inhibitors) supplied with 0.2% NP-40, and after 5 min, the axonemes were spun down (30 min at $23,300 \times g$ at 4 °C) and suspended in lysis buffer (50 mM Tris-HCl, pH 7.4, 0.4% SDS, 0.5 M NaCl, 1 mM DTT with protease inhibitors). After an hour, samples were spun down ($8000 \times g$ at 4 °C), and the supernatant was diluted with 50 mM Tris-HCl, pH 7.4 (1:3). The biotinylated proteins were incubated overnight with streptavidin-coupled Dynabeads (Dynabeads M-280 Streptavidin, Thermo Fisher Scientific, Waltham, MA) at 4 °C. The bead-bound biotinylated proteins were analyzed by Western blotting using HRP-streptavidin diluted 1:40 000 in 3% BSA/TBST (Thermo Fisher Scientific, Rockford, IL, USA) and identified by mass spectrometry.

Differential Quantitative Cilia Proteome Analyses

Sample Preparation

Cilia purified from 5×10^7 cells were resuspended in axoneme stabilization buffer containing 0.2% NP-40 and incubated for 5 min on ice to remove the ciliary membrane. Axonemes were lysed for 1 hour at RT with lysis buffer (1% SDS, 50 mM Tris-HCl, pH 7.4), and protein concentrations were determined by a Pierce™ BCA Protein Assay Kit (Thermo Scientific, Bartlesville, OK, USA). Three hundred micrograms of protein was precipitated using a ReadyPrep 2-D Cleanup Kit (Bio-Rad Laboratories, USA) at -20 °C overnight. Next, samples were centrifuged at $20,000 \times g$ for 30 min at 4 °C (5804/5804 R Centrifuge, Eppendorf, USA), and supernatants were discarded. Protein pellets were dissolved in 0.1% RapiGest and 500 mmol/L TEAB and then incubated at 850 rpm for 45 min at 37 °C (Eppendorf Comfort Thermomixer, Eppendorf, USA). Proteins were digested by trypsin (Trypsin Gold, Mass Spectrometry Grade, Promega; protein:enzyme (w/w) ratio – 100:1, at 37 °C for 16 h) according to a standard protein digestion protocol including reduction (by 1,4-dithiothreitol) and alkylation (by iodoacetamide). The digestion reaction was stopped by the addition of 55% trifluoroacetic acid (final concentration of 5%), samples were centrifuged at $20,000 \times g$ for 30 min

at 4 °C to precipitate RapiGest, and pellets were discarded. Supernatants containing the obtained peptides were purified using Pierce™ Peptide Desalting Spin Columns (Thermo Scientific, Bartlesville, OK, USA) according to the manufacturer's protocol, dried in a vacuum concentrator at RT (SpeedVac Concentrator Plus, Eppendorf, USA) and stored at -80 °C for further analysis.

LC-MS/MS analysis of labeled peptides (TMT analysis)

Desalted peptides were dissolved in 100 µl of 100 mmol/l TEAB solution, and peptide concentrations were determined using the Pierce™ Quantitative Fluorescent Peptide Assay (Thermo Scientific, USA). Next, a TMT labeling reaction was performed according to the procedure provided by the manufacturer. Briefly, a volume of sample containing 30 µg of peptides was labeled with a corresponding tandem mass tag. The labeling reaction was carried out for 1 h at room temperature and quenched with 5% hydroxylamine. Additionally, a test sample was prepared to check the efficiency of labeling. Labeled peptides were purified and fractionated using liquid chromatography at high pH. Separation was carried out for 26 minutes at a flow rate of 0.8 ml/min using a UPLC system (Acquity UPLC Class H system, Waters). The mobile phases consisted of water (A), acetonitrile (B) and 100 mmol/L ammonia solution (C). The percentage of phase C was kept constant at 10% throughout the separation. Fractions were collected every 1 minute, starting from the second minute of the run. The peptide elution was monitored spectrophotometrically at 214 nm. Twenty-four fractions were collected and combined to obtain 12 measurement samples. Samples were dried in a vacuum concentrator at room temperature. Peptides were resuspended in 100 µl of 5% acetonitrile and 0.1% formic acid.

Peptides were analyzed on the Evosep One system (Evosep Biosystems, Odense, Denmark) coupled to an Orbitrap Exploris 480 mass spectrometer (Thermo Fisher Scientific, USA) according to (Bekker-Jensen et al., 2020 [DOI](#)). One µg of peptides was loaded onto Evotips C18 trap columns (Evosep Biosystems, Odense, Denmark) according to the manufacturer's protocol with some modifications. Chromatographic separation of peptides was carried out using a mobile phase flow rate of 500 nl/min in gradient elution mode for 44 min on an EV1106 analytical column (Dr. Maisch C18 AQ, particle size 1.9 µm, 150 µm x 150 mm, Evosep Biosystems, Odense, Denmark). The following gradient elution was used: 0 min – 1% B, 120 min – 35% B, 121 min – 95% B, 124 min – 1% B, 127 min – 95% B, 130 min – 1% B. The eluted peptides were ionized in the positive ion mode in the nano-ESI source with a capillary voltage of 2.1 kV and the temperature of transfer capillary 275 °C. Survey scans from 300 *m/z* to 1700 *m/z* were acquired by an Orbitrap mass analyzer at a resolving power of 60 000. The resolving power in the MS2 spectrum measurement mode was 30 000 with the TurboTMT function set to TMT Reagents. HCD-MS/MS spectra (normalized collision energy of 30%) were generated for 25 multiply charged precursor ions from each survey scan. A precursor fit filter was applied to reduce peptide co-fragmentation. Dynamic exclusion was set to 20 s, and the precursor ion intensity threshold was set to 5×10^3 .

LC-MS/MS analysis of non-labeled peptides (label-free analysis)

Desalted peptides were resuspended in 100 µl of 5% acetonitrile and 0.1% formic acid. Next, peptide samples were analyzed on the Evosep One system (Evosep Biosystems, Odense, Denmark) coupled to an Orbitrap Exploris 480 mass spectrometer (Thermo Fisher Scientific, USA) according to (Bekker-Jensen et al., 2020 [DOI](#)) with some modifications. Chromatographic separation and mass spectrometry detection were conducted as described in a previous article (Bicka et al., 2022 [DOI](#)).

Data analysis

MS data were analyzed with FragPipe (v. 17.1) (Nesvilab, University of Michigan, Ann Arbor, MI) with MSFragger (v. 3.4) (Kong et al., 2017 [DOI](#)) and Philosopher (v. 4.2.1) (da Veiga Leprevost et al., 2020 [DOI](#)). ProteoWizard's MSConvert (v. 3.0.1908) (Palo Alto, CA) (Chambers et al., 2012 [DOI](#)) was used to convert the raw MS data to mzML format. The *Tetrahymena thermophila* UniProt database (canonical and isoform sequences; 27,027 entries) was searched using the following search

parameters: (i) digestion enzyme-trypsin/P, up to two missed cleavage sites were allowed, (ii) precursor and fragment ion mass tolerance ± 10.0 ppm and ± 20 ppm, respectively, (iii) fixed modifications: carbamidomethyl (C), (iv) variable modifications: oxidation (M), deamidation (N), (Q). For TMT samples, additional fixed modification was used (v) TMT modification of lysine (+229.16293) and variable modification (vi) TMT modification of N-termini of protein and peptide (+229.16293). Proteins and peptides were identified using the target-decoy approach with a reversed database. The peptide mass range was set from 500 Da to 5 000 Da. The results were processed with FDR (false discovery rate) set to 1% at the PSM, peptide and protein levels. Quantitative analysis was performed using IonQuant (label-free quantitation) and TMT Integrator (TMT-based quantitation). Statistical analysis was performed with Perseus (v. 2.0.3) (Max Planck Institute of Biochemistry, Martinsried, Germany). In label-free quantitation, missing values were replaced based on quantile regression imputation of left-censored data (QRILC) (Rudolph & Cox, 2019 [DOI](#)). Student T-test was used for statistical analysis. Proteins were considered to be differentially expressed if the difference in abundance was statistically significant (FDR adjusted p-value < 0.05) and the fold change was equal to or higher than 1.5.

Reagent availability

All plasmids and *Tetrahymena* mutants are available on request. LFQ and TMT mass spectrometry data are provided as **Figure 4** [DOI](#)-Source Data 1' and **Figure 4** [DOI](#)-Source Data 2'. The list of all protein identified in co-IP and BioID experiments are provided as **Figure 6** [DOI](#)-Source Data 1' and **Figure 6** [DOI](#)-Source Data 2'. Nucleotide sequences of the primers used to amplified PCR fragments are provided in Table S1.

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Competing interest

The authors declare that no competing interests exist.

Author contributions

Marta Bicka, Investigation, Validation, Formal analysis, Visualization Corbin Black, Investigation,

Avrin Ghanaeian, Formal analysis, Investigation, Writing – review & editing, Visualization Ewa Joachimiak, Methodology, Investigation, Validation, Formal analysis, Visualization, Funding acquisition, Writing—Review and Editing

Anna Osinka, Investigation, Visualization Sumita Majhi, Investigation,

Anna Konopka, Methodology, Validation, Resources, Supervision

Ewa Bulska, Supervision, Resources, funding acquisition, writing - review & editing

Khanh Huy Bui, Investigation, Supervision, Resources, Funding acquisition, Writing – review & editing

Dorota Wloga, Conceptualization, Investigation, Validation, Visualization, Resources, Formal analysis, Supervision, Funding acquisition, Project administration, Writing – original draft, Writing – review and editing

Supplementary Tables

Table S1. List of primers used in this study. The nucleotide sequences recognized by the restriction endonucleases are in bold.

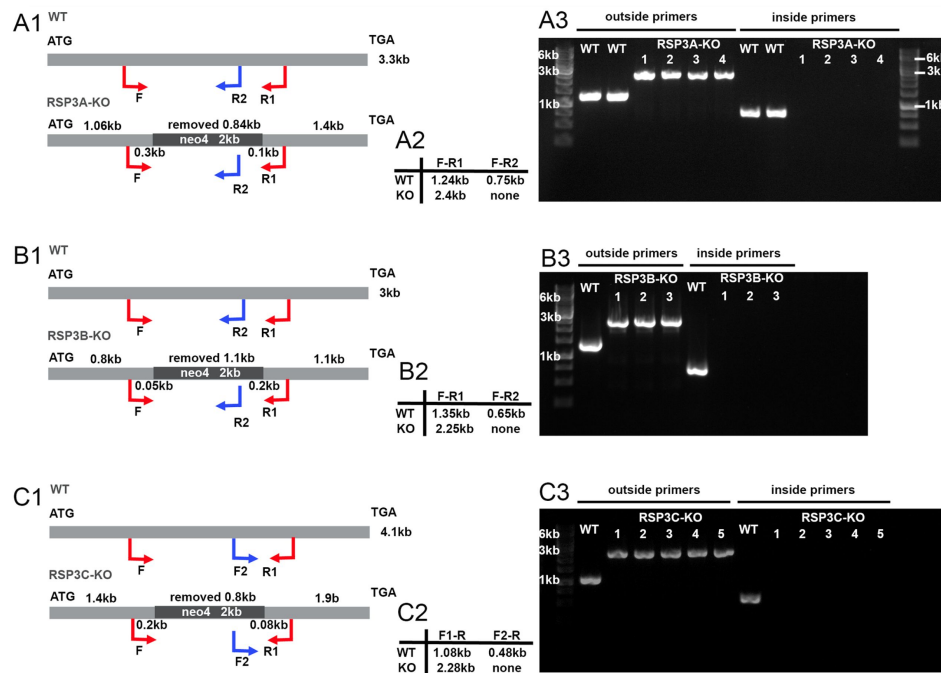
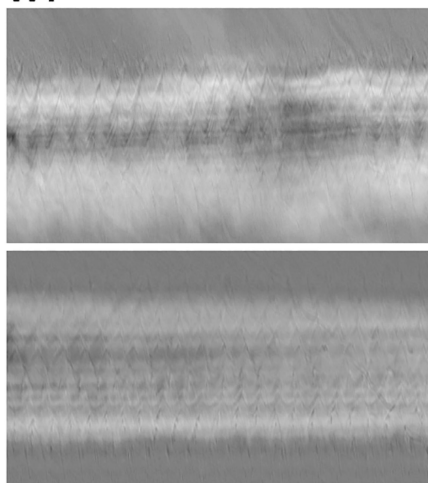


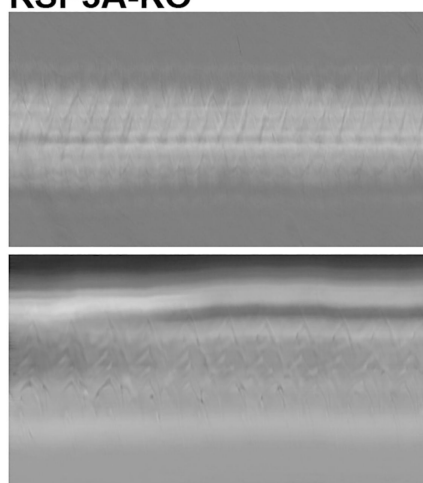
Figure 2-figure supplement 1.

Alterations in *RSP3A*, *RSP3B*, and *RSP3C* loci in engineered *Tetrahymena* knockout mutants. (A1, B1, C1) Schematic representations of the *RSP3A* (A), *RSP3B* (B), and *RSP3C* (C) loci in the wild-type genome (WT, upper gray bar) and obtained knockout cells (RSP3-KO, lower gray bar) with the size of the fragment of the open reading frame removed in the knockout cells indicated and the position of the annealing of primers (blue and red arrows) used to test changes in the analyzed loci marked. Numbers near the blue arrows indicate the distance of the primer annealing site from the neo4 cassette that replaced the removed fragment of the open reading frame in mutated loci. Note that the primer represented by a red arrow recognizes the nucleotide sequence deleted in mutant cells. Numbers above the gray bars representing the open reading frame indicate the distance between either the start or stop codons and the neo4 cassette insertion site. (A2, B2, C2) A table showing the expected size of the PCR products obtained using genomic DNA purified from wild-type (WT) and knockout cells (KO) and primers as indicated. (A3, B3, C3) PCR analyses of the *RSP3* loci *RSP3A* (A3), *RSP3B* (B3), and *RSP3C* (C3) in independently obtained *RSP3* mutants using either both primers annealing outside the neo4 cassette or one of the primers recognizing the deleted gene fragment as indicated in A1, B1, and C1, respectively.

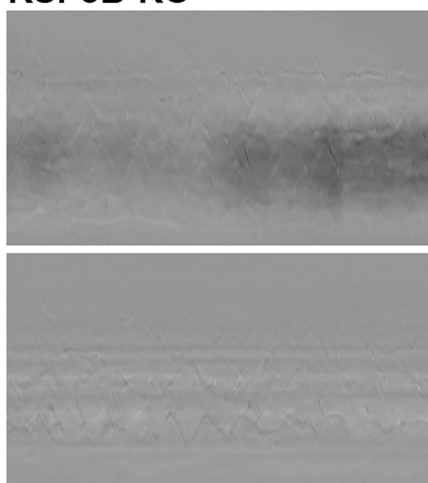
WT



RSP3A-KO



RSP3B-KO



RSP3C-KO

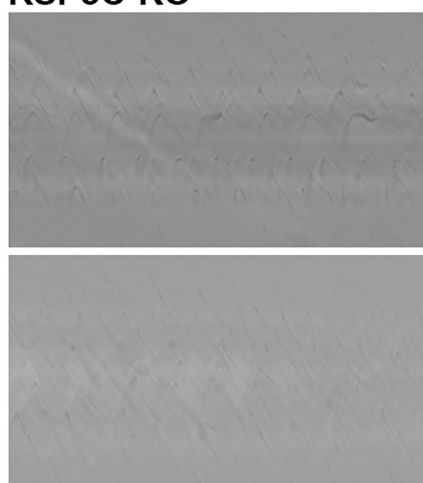


Figure 2-figure supplement 2.

Knockout of RSP3B affects cilia beating synchrony. Examples of the kymographs showing cilia beating synchrony.

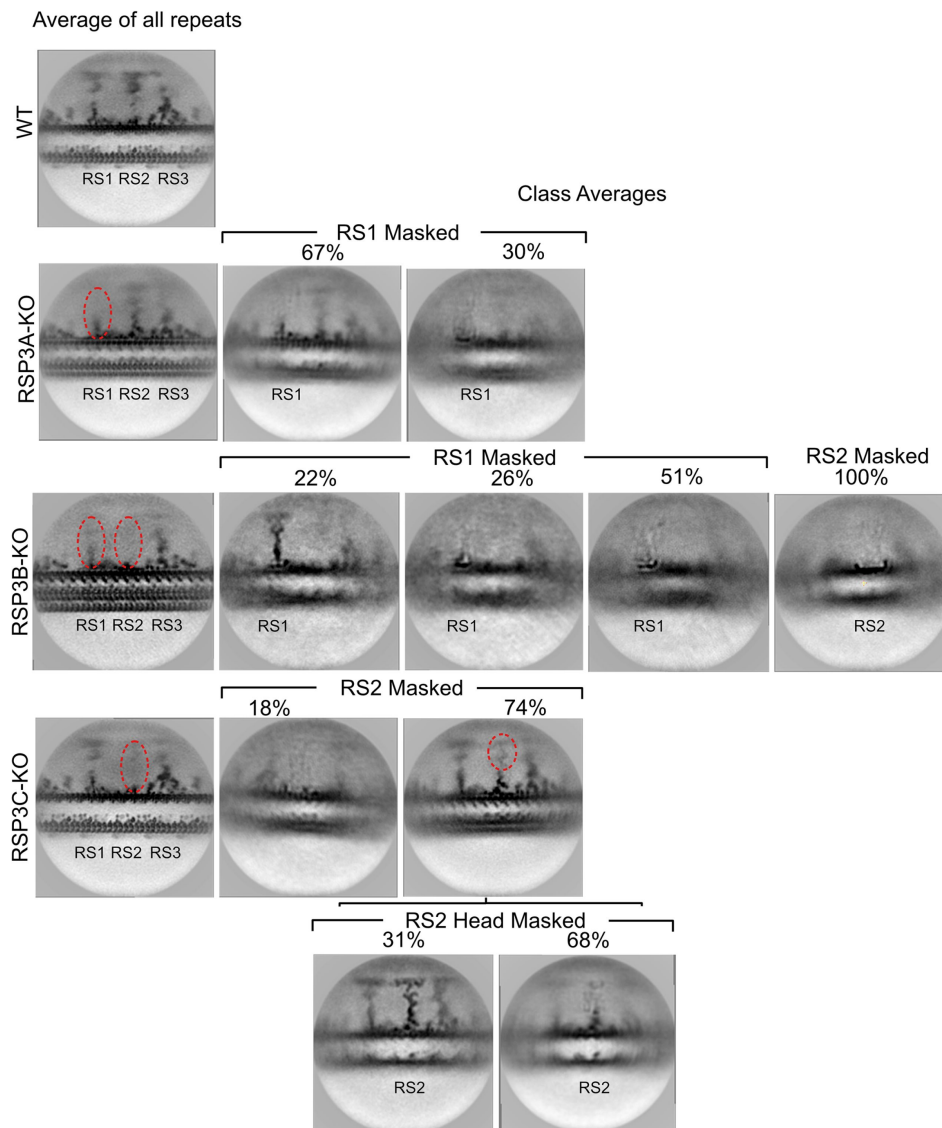


Figure 3-figure supplement 1.

Consensus subtomogram averages and three-dimensional classification of the radial spokes in *Tetrahymena* RSP3 knockout mutants. In RSP3A-KO mutant cilia, RS1 is missing in ~35% of 96-nm axonemal units. The analyses of n=2099 axonemal repeats with RS1 defects revealed either the lack of the entire RS1 (~30%, n=639 units) or RS1 except for the RS1 base (~67%, n=1417). Three percent of the recorded repeats were not classified due to the low number of particles. In RSP3B-KO mutant cilia, all analyzed axonemal repeats (n=2092) lacked RS2. Additionally, 77.5% of the axonemal repeats (n=1622) had RS1 spokes defects. The RS1 spokes were either completely missing (51.5%, n=1078) or only their base was well visible (26%, n=544). Among the collected n=2790 axonemal repeats of RSP3C-KO mutant cilia, 173 axonemal repeats (7%) were unclassified. The analyses of the remaining n=2617 axonemal repeats revealed that ~20% (n=527) lacked the entire RS2. The 3D classification of the remaining 80% of repeats using the RS2 mask covering an RS2 head and stalk showed that the remaining units grouped into two categories: (i) with an intact RS2 structure (~18%, n=469) and (ii) with a well-visible RS2 base (~62%, n=1621).

Method	Subtomogram Averaging			
Dataset	<i>WT</i>	<i>RSP3A-KO</i>	<i>RSP3B-KO</i>	<i>RSP3C-KO</i>
Microscope	Titan Krios	Titan Krios	Titan Krios	Titan Krios
Electron Detector	Gatan K3	Gatan K3	Gatan K3	Gatan K3
Zero-loss filter (eV)	20	20	20	20
Magnification	42,000	42,000	42,000	42,000
Voltage (keV)	300	300	300	300
Exposure (e/Å ²)	160	160	80	160
Defocus (μm)	1.0-3.0	3.5-5.0	3-4	2.5-4.0
Pixel size (Å)	2.12	2.12	2.12	2.12
Tilt range (increment)	-60° - 60° (3°)	-60° - 60° (3°)	-60° - 60° (3°)	-60° - 60° (3°)
Tilt scheme	dose symmetric	dose symmetric	dose symmetric	dose symmetric
Tilt series acquired	58	37	50	38
Subtomo averaged	2608	2099	2093	2790
Symmetry imposed	C1	C1	C1	C1
Map resolution (Å)	19	20	22	17

Figure 3-figure supplement 2.

Data collection and analysis parameters for cryo-ET and subtomogram averaging.

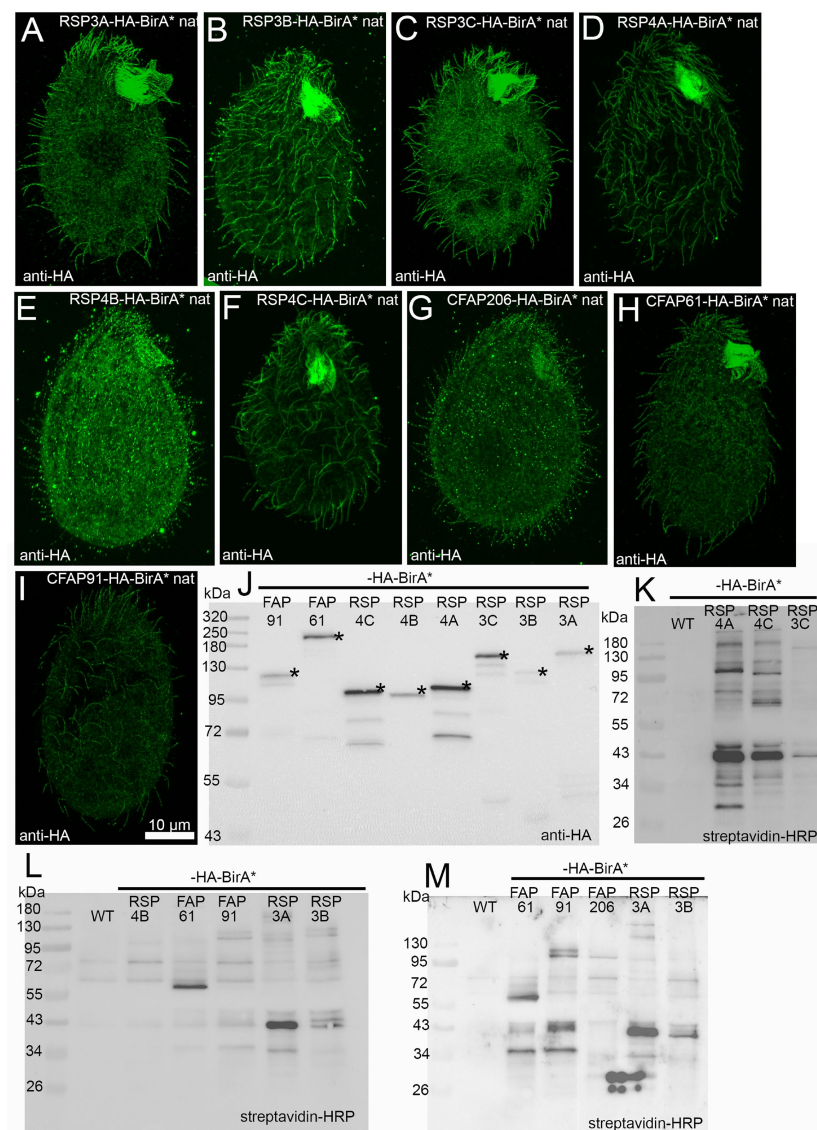


Figure 6-figure supplement 1.

(A-I) Immunofluorescence confocal images of *Tetrahymena* cells expressing RSP proteins as C-terminal -HA-BirA* fusions under the control of respective native promoters detected using anti-HA antibodies. (A) RSP3A-HA-BirA*, (B) RSP3B-HA-BirA*, (C) RSP3C-HA-BirA*, (D) RSP4A-HA-BirA*, (E) RSP4B-HA-BirA*, (F) RSP4C-HA-BirA*, (G) CFAP206-HA-BirA*, (H) CFAP61-HA-BirA*, (I) CFAP91-HA-BirA*. (J-M) Western blot analyses of the ciliary proteins isolated from *Tetrahymena* cells expressing RSP proteins as -HA-BirA* fusions under the control of respective native promoters. A star indicates a band corresponding to the position of fusion proteins. (J) Western blot-based identification of the RSP-HA-BirA* fusion proteins. (K-M) Western blot-based analyses of the biotinylated ciliary proteins in either WT cells or cells expressing RSP-HA-BirA* under the control of the respective native promoter, all grown for 4 hrs in a 10 mM Tris-HCl buffer (pH 7.4) supplemented with biotin, detected using HRP-conjugated streptavidin.

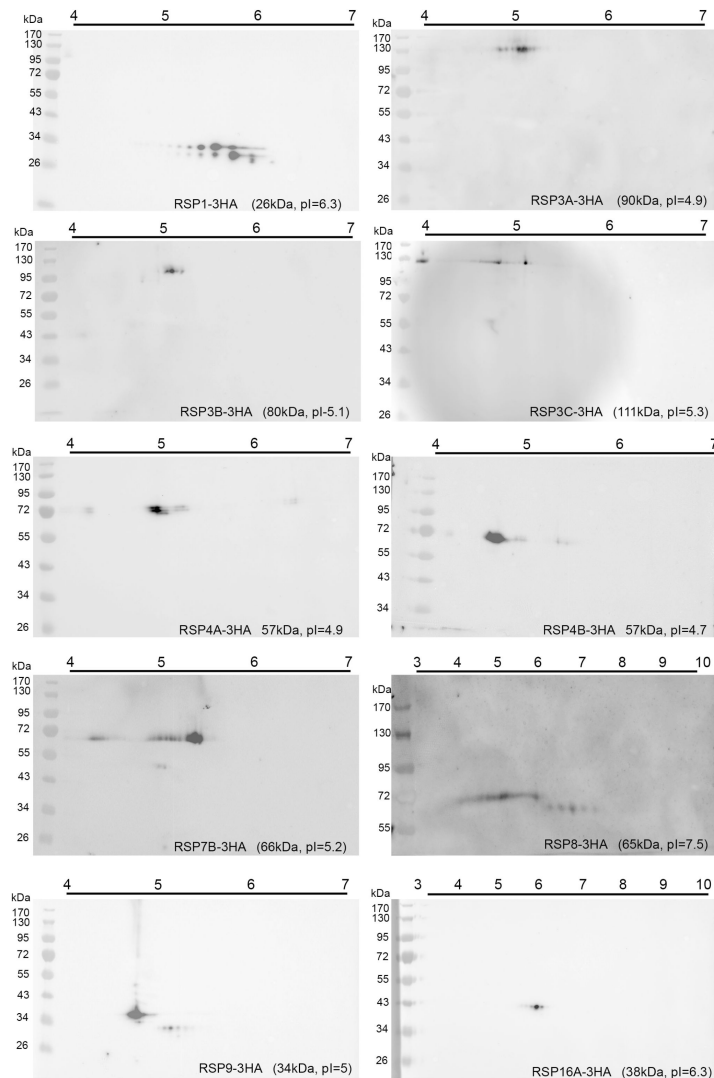


Figure 6-figure supplement 2.

Two-dimensional analyses of the RSP isoforms. The radial spokes proteins were expressed as C-terminal 3HA fusions under the control of the respective native proteins. The isoelectric focusing of the ciliary proteins (30 μ g) was performed using 7 cm 4-7 or 3-10 ready strips. The theoretical pI and Mw values were calculated using https://web.expasy.org/compute_pi/. The RSP isoforms were detected using anti-HA antibodies.

RS protein	RS part	domains	MW (kD)	pI	TGD number THERM_	% of identity / similarity	E-value (TGD blastp)
RSP1	head	MORN	26	6.3	THERM_000196029	31/49	3.19514e-12
RSP2	neck	DPY-30 CC	75	4.7	THERM_00394410	37/51 within DPY	0.000540192
RSP3	stalk	AKAP AKAP ARF, AKAP	90 80 111	4.9 5.1 5.3	RSP3A THERM_01044600 RSP3B THERM_00566810 RSP3C THERM_00418270	44/60 41/65 31/54	2.34814e-19 2.1949e-41 8.00525e-20
RSP4	head	Unknown	57	4.9	RSP4A THERM_00427590	23/39	1.40043e-13
RSP6		Unknown	57	4.7	RSP4B THERM_00502580	22/43	5.25581e-26
		Unknown	54	4.9	RSP4C THERM_00444180	30/48	2.00863e-31
RSP5					not found		
RSP7	stalk	RIIa, IQ	39	4.7	RSP7A THERM_00670610	50/64 (N-term only)	3.31578e-05
		AKAP28, EF hand	66	5.2	RSP7B THERM_00492840	20/38	6.65092e-05
RSP8	stalk	ARM-like	65	7.5	THERM_00313520	30/47	4.89747e-16
RSP9	head	unknown	34	5	THERM_00430020	26/43	6.28284e-12
RSP10	head	MORN	27	5.3	THERM_00378600	36/49	1.78444e-19
RSP11	stalk	RIIa	9	7.9	THERM_00540090	60/72	1.62388e-11
RSP12	stalk near neck	Pro-isomerase Fructose-1,6-bisphosphatase	126	9.3	RSP12A THERM_01018330	37/58 (cover 11%)	2.06701e-19
RSP12	stalk near neck	Pro-isomerase Fructose-1,6-bisphosphatase	102	9.0	RSP12B THERM_00466090	33/49	3.9832e-18
RSP13					no Chlamydomonas sequence available		
RSP14	stalk	RPT2, HEAT ARM-like	41	5.7	THERM_000773538	20/43	0.481103
RSP15	stalk	LRR	47	5.7	LRRC34 THERM_00695730	28/46	5.12592e-19
RSP16	neck	DNAJ/DNAJ-C	38	6.3	RSP16A THERM_00471260	36/53	1.2198e-43
		DNAJ/DNAJ-C	39	6.2	RSP16B THERM_00238810	32/51	3.53246e-40
RSP17					not found		
RSP18	RS2,	FAP91 domain	76	7.3	CFAP91 THERM_00578560	33/53	2.20388e-19



Figure 6-figure supplement 3.

Identification of *Tetrahymena* RSP orthologs

Figure 6-figure supplement 3-Table 1A. List of *Tetrahymena thermophila* radial spoke proteins identified in *Tetrahymena* Genome Database (TGD, <https://tet.ciliate.org>) using *Chlamydomonas* RSP proteins as baits. In the case of multiple hits (see details in **Table S1B**), it was analyzed if identified *Tetrahymena* RSP orthologs are present in the *Tetrahymena* ciliomes (**Figure 4** -Source Data 2'), co-immunoprecipitate with known RSPs (**Figure 6** -Source Data 1'), or are biotinylated in cilia of cells expressing RSP proteins fused with mutated BirA* ligase (**Figure 6** -Source Data 2'). The identification of *Tetrahymena* orthologous proteins was verified by a reverse blast search against *Chlamydomonas reinhardtii* proteins (<https://blast.ncbi.nlm.nih.gov>). Domains were identified using SMART (<http://smart.embl-heidelberg.de/>) and InterProScan (<https://www.ebi.ac.uk/interpro>) programs.

Figure 6-figure supplement 3-Table 1B. *Tetrahymena* RSP orthologs identified with multiple hits in TGD. In cases when numerous proteins were identified with similar scores in blastp search of *Tetrahymena* total proteome (TGD) using *Chlamydomonas* RSP as a bait, it was analyzed which of those proteins are present in *Tetrahymena* ciliome (Tables S3 and S4) and either co-immunoprecipitate (co-IP, Table S5) with known RSPs or are biotinylated (BioID, Table S6) in cells expressing known RS proteins as fusions with mutated BirA* ligase. Arrows indicate if there was a significant change (↑ increase or ↓ decrease) in the protein level in studied knock-out mutants (based on TMT analyses, Table S4). ns – change not significant, nd – protein not detected in the *Tetrahymena* ciliome.

Figure 6-figure supplement 3-Table 2. Comparative analyses of the levels of the radial spoke proteins in wild-type cells and radial spoke mutants

Figure 6-figure supplement 3-Table 3. Comparative analyses of the levels of potential radial spoke-associated proteins in wild-type cells and radial spoke mutants

Figure 6-figure supplement 3-Table 4. Comparative analyses of the levels of proteins with an enzymatic activities in radial spoke mutants

	RS3 adaptor	CC					
RSP19	RS3 stalk	DUF4821 FAD/NAD(P) binding CC	198	5.4	CFAP61 T THERM_00641200	25/42	4.98635e-36
RSP20	stalk	EF hand	17	4	Calmodulin T THERM_00630500	88/95	7.17305e-92
RSP20 (?)	stalk?	EF hand	24	4.8	CaM 7-2 T THERM_00549500	25/54	4.0036e-14
RSP21		no Chlamydomonas sequence available					
RSP22	stalk	LC8	10	6.8	T THERM_000649439	85/92	7.13575e-57
RSP23 CFAP67B	neck	nucleoside diphosphate kinase 6	43	6.2	T THERM_000372529	49/61	8.23531e-33
FAP206	RS2 adaptor	FAP206 domain	74	6.03	T THERM_00820660	33/53	1.47723e-10
FAP251	RS3 base	WD40, EF hand	114	4.8	T THERM_01262850	29/50	1.27034e-122
FAP198	stalk	Cyt-b5	28	5.1	FAP198A T THERM_01092470	49/69	1.3317e-53
		Cyt-b5	28	5.1	FAP198B T THERM_00476710	47/65	1.872e-48
FAP207	RS1, RS2 base	MORN	48	4.9	T THERM_00529880	41/52	2.76951e-48
FAP253/ IQUB/ CMUB116	RS1 base	IQ	49	9	T THERM_00316930	52/69	4.36935e-87

AKAP - A-kinase anchoring protein; ARF - ADP-ribosylation factor; ARM - armadillo repeat; CC - coiled coil; Cyt-b5 - Cytochrome b5; DNAJ - Hsp40 type protein; DPY-30 - domain found in the Dpy-30 proteins; DUF - domain of unknown function; EF hand - calcium-binding; HEAT - tandemly repeated, amino acid module present in huntingtin, elongation factor 3, alpha regulatory subunit of PP2A, and TOR1; IQ - calmodulin-binding motif; LRR - leucine-rich repeat; MORN - Membrane Occupation and Recognition Nexus motif; RIIa - Regulatory subunit portion of type II PKA R-subunit; WD40 - WD-40 repeats;

		E-value (TGD blastp)	In TMT cilome	In co-IP or BioID	RSP3A-KO	RSP3B-KO	RSP3C-KO	FAP91-KO
RSP1	T THERM_000196029	1.78444e-19	yes	coIP 3A (1) Bir 3A, 3B, 4A, 4C	ns	↓	ns	ns
RSP10	T THERM_00378600	1.90468e-19	yes	coIP 4A (5/5) Bir 3A, 3B, 4A, 4C	ns	↓	ns	ns
	T THERM_00338260	7.69708e-19	no	-	-	-	-	-
	T THERM_00561720	4.08366e-17	yes	nd	ns	ns	ns	ns
	T THERM_00086700	3.34742e-16	yes	Bir 3B (1), 4A (1)	ns	↓	↓	↓
	T THERM_00940320	4.65599e-16	no	-	-	-	-	-
	T THERM_00471440	3.45586e-15	no	-	-	-	-	-
	T THERM_00522150	6.02902e-15	no	-	-	-	-	-
	T THERM_00013150	6.59314e-15	no	-	-	-	-	-
	T THERM_00122390	1.03558e-14	no	-	-	-	-	-
	T THERM_01044720	4.35786e-14	no	-	-	-	-	-
	T THERM_00058629	8.15103e-14	yes	nd	ns	↓	ns	ns
	T THERM_00047490	8.87354e-14	yes	nd	↑	ns	ns	↓
	T THERM_00080020	1.83831e-13	yes	nd	ns	ns	ns	ns
	T THERM_00138540	1.85468e-13	no	-	-	-	-	-
	T THERM_00133460	4.62151e-13	no	-	-	-	-	-
	T THERM_00149910	5.33197e-13	no	-	-	-	-	-
	T THERM_00691730	6.19233e-13	no	-	-	-	-	-
	T THERM_00142460	6.24492e-13	no	-	-	-	-	-
	T THERM_01248870	6.40885e-13	no	-	-	-	-	-
	T THERM_00724730	1.06173e-12	no	-	-	-	-	-
	T THERM_00078940	1.13884e-12	yes	nd	ns	↓	ns	ns
	T THERM_00685930	1.18943e-12	yes	nd	ns	↓	↓	↓
	T THERM_00599110	2.45575e-12	yes	nd	ns	ns	↑	ns
	T THERM_00486410	4.33653e-12	yes	nd	ns	ns	ns	ns
	T THERM_00590130	5.60043e-12	yes	nd	ns	↓	↓	↓
	T THERM_00077350	6.44267e-12	yes	nd	ns	ns	ns	ns
	T THERM_00329860	6.74109e-12	no	-	-	-	-	-
	T THERM_01084140	8.55508e-12	no	-	-	-	-	-
	T THERM_00637100	8.74526e-12	no	-	-	-	-	-
	T THERM_00196080	1.13797e-11	yes	nd	↓	↓	↓	↓
	T THERM_00766490	1.31424e-10	yes	nd	↓	↓	↓	↓
	T THERM_00584930	1.95024e-10	yes	nd	↑	↑	ns	↓
	T THERM_00732890	3.04534e-10	yes	nd	ns	ns	ns	ns
RSP5	T THERM_00962090	none	yes	nd	↑	ns	↑	↑
	T THERM_00338200	none	yes	nd	↑	ns	ns	↓
	T THERM_01080330	none	yes	nd	↑	↑	ns	ns
	T THERM_00058260	none	yes	nd	↑	↑	ns	ns
	T THERM_00526310	none	yes	nd	↑	↑	↑	ns
	T THERM_00300640	none	yes	nd	↑	ns	ns	ns
	T THERM_00300630	none	yes	nd	ns	↓	↓	ns
	T THERM_00526449	none	yes	nd	ns	ns	↑	↓
	T THERM_00274520	none	yes	nd	ns	↓	↓	↓
	T THERM_00266330	none	yes	nd	ns	↓	ns	ns
	T THERM_00300620	none	yes	nd	ns	ns	ns	↑

Figure 6-figure supplement 3. (continued)

	TTHERM_00703880	none	yes	nd	ns	↑	↑	ns
	TTHERM_01002600	none	yes	nd	ns	ns	↓	↓
RSP7	TTHERM_00353490	7.66049e-13	yes	nd	ns	↑	ns	↑
	TTHERM_00686010	2.22411e-09	no	-	-	-	-	-
	TTHERM_00384910	5.51363e-09	yes	nd	↓	ns	ns	↑
	TTHERM_00632950	1.57971e-08	no	-	-	-	-	-
	TTHERM_00486220	8.14257e-08	no	-	-	-	-	↑
	TTHERM_00523059	1.054e-07	yes	nd	↓	ns	↓	-
	TTHERM_00059190	1.12651e-07	yes	nd	↓	↓	↓	↓
	TTHERM_00668030	2.30983e-07	yes	nd	ns	ns	ns	ns
	TTHERM_000442899 (STK36-like)	4.65608e-07	yes	co-IP RSP4C (1) Bir RSP4A (1-2), 3B (1) 4B (3/3)	ns	↑	ns	↑
	TTHERM_00300120	6.8876e-07	yes	nd	ns	↑	ns	ns
	TTHERM_00630500 (CAMI)	9.98653e-07	yes	Bir 4A (2-3), 3B (1- 11) F206 (1)	ns	↓	↓	ns
	TTHERM_00622900	2.83945e-06	no	-	-	-	-	-
	TTHERM_00267880 (BBC37)	4.07889e-06	yes	coIP 3A(1), 4A(4)	ns	↓	↓	↑
	TTHERM_00194410	5.09536e-06	no	-	-	-	-	-
	TTHERM_00616570	5.2479e-06	yes	nd	↓	ns	ns	ns
	TTHERM_00218310	9.61136e-06	yes	nd	ns	↓	↓	↓
	TTHERM_00589960	1.36054e-05	yes	nd	ns	ns	ns	ns
	TTHERM_00899430	- 1.57e-05	no	-	-	-	-	-
	TTHERM_00495950	1.90483e-05	yes	nd	↑	ns	ns	↑
	TTHERM_01052880	2.2429e-05	yes	nd	ns	↓	ns	↑
	TTHERM_00670610 (RSP7A)	3.31578e-05	yes	co-IP 4A(2), 4B(2), 4C(1), 3A(8) Bir 3A(1-5), 3B(4- 13), 3C(11), 4A(14- 16), 4C(1-3), 206 (4)	ns	↓	ns	↓
	TTHERM_00637390	4.33175e-05	yes	nd	↑	↓	ns	↓
	TTHERM_000841299	5.08855e-05	no	-	-	-	-	-
	TTHERM_00147620	6.24849e-05	yes	nd	ns	ns	ns	ns
	TTHERM_00492840 (RSP7B)	6.65092e-05	yes	coIP 3A (4), 4A(4), 4B (1) Bir 3B(2), 3C(1), 4A(1), 206(1), 61(3- 5)	↑	↑	↑	↓
	TTHERM_00136380	0.00013603	no	-	-	-	-	-
	TTHERM_00079490	0.00017402	yes	nd	ns	↑	ns	ns
	TTHERM_00013440	0.00017893	no	-	-	-	-	-
	TTHERM_00371010	0.00021959	yes	nd	ns	↓	ns	ns
RSP12	TTHERM_00051740	1.0883e-25	yes	Bir 3 A(1 pep)	ns	↑	ns	ns
	TTHERM_00997720	1.64934e-25	no	-	-	-	-	-
	TTHERM_00890120	2.17043e-25	yes	nd	↑	ns	↑	↑
	TTHERM_00243800	1.07131e-22	no	-	-	-	-	-
	TTHERM_00118710	5.99925e-22	yes	nd	ns	↓	↓	ns
	TTHERM_00492720	6.01654e-22	no	-	-	-	-	-
	TTHERM_00548130	4.03513e-21	yes	nd	ns	↓	↓	↑
	TTHERM_01018330	2.06701e-19	yes	coIP 3a (1) BioID4B (4)	ns	↓	↓	↓
	TTHERM_00865350	5.9945e-19	yes	nd	↑	↓	ns	ns
	TTHERM_00877050	1.9254e-18	no	-	-	-	-	-
	TTHERM_00466090	3.9832e-18	yes	Bir 4a (1)	↓	↓	ns	ns
	TTHERM_00075590	7.37871e-18	no	-	-	-	-	-
	TTHERM_00772040	1.51296e-17	no	-	-	-	-	-
	TTHERM_00538480	7.1862e-15	no	-	-	-	-	-
	TTHERM_00757550	0.00121392	no	-	-	-	-	-
	TTHERM_00770670	0.82296	yes	coIP 3A (5/2) 4A (4/4) 4B (2/2) 4C (1/1) FAP251(1-2/1-2)	ns	ns	ns	↓
RSP22	TTHERM_000649439	7.13575e-57	yes	coIP 3A (11/5) Bir 3A, 3B, 4A, 4C	ns	↓	ns	ns
	TTHERM_000463528	2.6695e-36	yes	nd	ns	ns	ns	ns
	TTHERM_00449029	7.66385e-31	yes	Bir (1/1)	ns	ns	ns	↓
	TTHERM_000756449	4.69204e-27	yes	nd	ns	ns	ns	ns
	TTHERM_00971840	2.90049e-24	yes	Bir (1/1)	ns	ns	ns	ns
	TTHERM_01079060	8.79124e-21	yes	nd	↓	ns	ns	↑
	TTHERM_00716250	3.10027e-13	yes	Bir (1/1)	ns	ns	ns	ns
	TTHERM_00023950	2.90337e-11	yes	Bir (1/1)	ns	ns	ns	↑
	TTHERM_000442909	5.67464e-07	yes	(1/1) coIP, Bir	ns	ns	ns	ns
FAP207	TTHERM_00529880	2.76951e-48	yes	coIP 3A, 4A, 4B, BioID 3A, 4A	ns	↓	ns	↓
	TTHERM_00584930	1.54591e-31	yes	nd	↑	↑	ns	↓
	TTHERM_00047490	1.45891e-27	yes	nd	↑	ns	ns	↓

RSP1 and **RSP10** – proteins contain multiple MORN domains. Using *Chlamydomonas* RSP1 as a bait, *Tetrahymena* RSP1 and RSP10 were identified as a 9th and 36th hits, using *Chlamydomonas* RSP10 as a bait, *Tetrahymena* RSP1 and RSP10 were identified as a 1st and 2nd hits (shown in the Table S1B). The MORN domain-containing **TTHERM_00086700** was reduced in mutants with *RSP3B*, *RSP3C*, or *CFAP91* deletion but was identified among biotinylated proteins only by a single peptide.

RSP5 has a **aldo-keto reductase** domain. The orthologous proteins were not identified in *Tetrahymena* proteome. Table S1B shows analyses of other (non-homologous) proteins described as aldo-keto reductases present in *Tetrahymena* cilium.

RSP7 – predicted *Tetrahymena* RSP11 is significantly smaller than *Chlamydomonas* paralog, similarity was detected only within N-terminal proteins' region.

RSP12 – **Peptidyl-prolyl cis-trans isomerase**. 8th hit (TTHERM_01018330) has the higher similarity in reverse blastp to *Chlamydomonas* RSP12. Protein TTHERM_00770670 highly reduced in CFAP91-KO cilia, was found in co-IP and BirA* assays but has no similarity to RSP12 in reverse blastp

Figure 6-figure supplement 3. (continued)

Radial spoke proteins	Acc number in TGD	co-IP	BioID	RSP3A-KO/WT				RSP3B-KO/WT				RSP3C-KO/WT				CFAP206-KO/WT		CFAP61-KO/WT		CFAP91-KO/WT					
				LQF		TMT		LQF		TMT		LQF		TMT		LQF		LQF		LQF		LQF		LQF	
				q-value	difference	q-value	difference	q-value	difference	q-value	difference	q-value	difference	q-value	difference	q-value	difference	q-value	difference	q-value	difference	q-value	difference	q-value	difference
RSP1	TTHERM_000196029	yes	yes	0.7795	-0.15	0.2168	-0.13	0.0171	-1.54	0.0016	-0.64	0.8424	-0.12	0.5512	-0.07	0.3602	-0.37	0.9789	-0.01	0.6253	0.21	0.1372	-0.13		
RSP2	TTHERM_00394410	yes	yes	0.4621	-0.50	0.0414	-0.29	0.0196	-1.54	0.0012	-0.59	0.3111	-0.74	0.3073	-0.13	0.3215	-0.42	0.4980	0.24	0.6309	-0.21	0.5120	-0.05		
RSP3A	TTHERM_01044600	yes	yes	0.0025	-5.23	0.0124	-0.99	0.0748	-1.12	0.0029	-0.40	0.7104	-0.22	0.6405	0.05	0.3774	-0.41	0.9426	0.02	0.8836	-0.08	0.0548	0.18		
RSP3B	TTHERM_00566810	yes	yes	0.4992	0.41	0.3772	-0.09	0.0010	-5.10	0.0019	-1.15	0.6048	0.30	0.8883	0.02	0.0768	-1.05	0.5058	-0.24	0.0040	-2.61	0.0041	-0.43		
RSP3C	TTHERM_00418270	yes	yes	0.9072	-0.06	0.8769	0.02	0.2197	-0.53	0.1760	-0.10	0.0009	-5.47	0.0044	-0.59	0.0085	-3.39	0.3490	0.43	0.0554	-1.69	0.0399	-0.21		
RSP4A	TTHERM_00247590	yes	yes	0.7013	-0.20	0.3759	-0.08	0.0320	-1.32	0.0011	-0.49	0.9359	-0.05	0.4944	0.07	0.1033	-0.83	0.7104	0.12	0.2889	-0.40	0.2864	-0.08		
RSP4B	TTHERM_00502580	yes	yes	0.3863	-0.80	0.9895	0.002	0.2212	-0.75	0.9626	0.00	0.0082	-4.58	0.0125	-0.79	0.0954	-0.94	0.9786	0.01	0.0484	-2.05	0.0219	-0.26		
RSP4C	TTHERM_00444180	yes	yes	0.3744	-0.74	0.1361	-0.18	0.0003	-3.62	0.0017	-0.68	0.5952	-0.27	0.5776	0.07	0.9839	-0.01	0.9724	0.01	0.8082	-0.11	0.6438	-0.03		
RSP7B AKA28_EF hand	TTHERM_00492840	yes	yes	0.5596	-0.30	0.0471	0.25	0.7444	-0.16	0.0031	0.35	0.6300	-0.27	0.0271	0.36	0.0815	-3.00	0.3705	-0.42	0.0043	-5.51	0.0020	-0.51		
RSP7A RIIAD11Q	TTHERM_00670610	yes	yes	0.4480	0.61	0.0700	-0.29	0.0441	-2.04	0.0019	-0.77	0.4556	0.57	0.8805	0.02	0.0963	-1.34	0.8033	-0.08	0.0248	-1.16	0.0044	-0.48		
RSP8	TTHERM_00313520	yes	yes	0.0444	-2.31	0.2889	-0.11	0.0180	-4.38	0.0006	-1.34	0.0040	-3.86	0.0017	-0.80	0.0247	-2.51	0.0381	1.12	0.0056	-3.19	0.0006	-1.21		
RSP9	TTHERM_00430020	yes	yes	0.7701	-0.15	0.4773	-0.08	0.0190	-1.59	0.0024	-0.40	0.6139	-0.27	0.1883	0.16	0.4841	-0.30	0.3612	-0.37	0.1901	-0.57	0.5680	0.04		
RSP10	TTHERM_00378600	yes	yes	0.3694	-0.78	0.4403	-0.08	0.0148	-2.16	0.0012	-0.58	0.3030	-0.79	0.7289	-0.04	0.5293	-0.23	0.7611	0.10	0.6808	-0.18	0.2148	-0.11		
RSP11	TTHERM_00540090	yes	yes	0.2115	-1.53	0.0307	-0.31	0.0111	-4.64	0.0011	-0.49	0.0869	-2.04	0.0367	-0.35	0.5919	1.14	0.1085	2.76	0.0309	1.01	0.0548	-0.20		
RSP12A	TTHERM_01018330	no	no	0.9256	0.05	0.2392	0.11	0.0161	-2.27	0.0079	-0.29	0.0010	-4.46	0.0118	-0.45	0.0096	-2.20	0.5560	-0.26	0.6036	-3.28	0.0019	-0.53		
RSP12B	TTHERM_00466090	no	yes	nd		0.0102	-0.48	nd		0.0000	-0.68	nd		0.0807	0.24	nd		nd		nd		0.5789	0.04		
RSP14	TTHERM_000773538	yes	yes	0.0377	-2.51	0.0045	-0.74	0.0146	-2.40	0.0010	-0.52	0.3733	-0.65	0.8355	0.02	0.6305	-0.24	0.1018	0.72	0.0130	1.08	0.0172	0.31		
RSP15LRR34	TTHERM_00695730	yes	yes	0.2194	-1.21	0.5601	-0.06	0.0120	-3.67	0.0019	-0.86	0.1759	-1.49	0.2121	-0.17	0.0080	-2.91	0.3356	-0.88	0.0042	-5.13	0.0038	-0.97		
RSP16A	TTHERM_00471260	yes	yes	0.0406	-2.62	0.0253	-0.33	0.0216	-2.25	0.0013	-0.56	0.0968	-1.59	0.5844	-0.07	0.3365	-0.51	0.2292	0.54	0.4597	0.35	0.4636	0.05		
RSP16B	TTHERM_00238810	yes	yes	0.8796	-0.08	0.3494	0.10	0.3395	-0.40	0.6381	0.03	0.0121	-5.13	0.0064	-0.67	0.0092	-1.92	0.8685	-0.05	0.0078	-1.24	0.0354	-0.22		
RSP20CaM1	TTHERM_00630900	no	no	nd		0.1719	-0.20	nd		0.0240	-0.32	nd		0.0117	-0.67	0.1659	-2.02	0.0406	1.00	0.0049	-3.10	0.3203	0.10		
calmodulin 7-2	TTHERM_00549500	yes	yes	0.0484	-2.36	0.0186	-0.38	0.1804	-0.83	0.3745	-0.06	0.0061	-4.45	0.0130	-0.44	0.6666	-0.22	0.9592	0.02	0.8873	-0.08	0.1067	0.14		
RSP22LC8	TTHERM_000649439	yes	yes	0.0666	-2.53	0.4513	-0.08	0.0439	-1.62	0.0027	-0.42	0.0461	-3.36	0.2872	0.11	0.0952	-1.00	0.2773	0.5	0.4440	-0.38	0.2457	0.08		
RSP23BFAF67B	TTHERM_000372529	yes	yes	0.6087	-0.29	0.9553	-0.01	0.7482	-0.16	0.6381	0.03	0.6874	-0.26	0.6799	0.04	0.5172	-0.25	0.5812	0.19	0.9600	0.03	0.0188	0.27		
RSP19FAF61	TTHERM_00641200	yes	yes	0.5334	-0.39	0.0010	1.32	0.5872	-0.23	0.0002	0.14	0.5468	-0.36	0.0044	1.46	0.6809	-0.16	0.0128	-3.60	0.0044	-3.56	0.1182	0.13		
FAF251	TTHERM_01262850	yes	yes	0.7834	-0.15	0.3688	0.09	0.6187	-0.23	0.4797	0.04	0.9708	-0.02	0.0437	0.28	0.3754	-0.34	0.9176	-0.03	0.0049	-5.61	0.0095	-0.88		
RSP18FAF91	TTHERM_00578360	yes	yes	0.6777	-0.21	0.3425	0.10	0.2540	-0.54	0.1045	-0.13	0.5640	-0.34	0.2188	0.14	0.3221	-0.40	0.7473	-0.10	0.0035	-5.85	0.0089	-0.97		
FAF206	TTHERM_00820660	yes	yes	0.8793	-0.08	0.4950	0.07	0.0008	-4.04	0.0074	-1.04	0.6660	-0.22	0.3175	-0.16	0.0139	-4.67	0.2723	-0.46	0.0041	-4.55	0.0083	-1.37		
FAF253	TTHERM_00316930	yes	yes	0.6415	-0.24	0.3314	-0.09	0.8457	-0.09	0.1084	0.12	0.5416	-0.36	0.4070	0.09	0.5152	-0.45	0.3267	0.55	0.6564	0.27	0.0557	0.18		
FAF207	TTHERM_00822230	yes	yes	0.8223	-0.12	0.5520	0.06	0.0137	-3.31	0.0040	-0.90	0.5665	-0.32	0.9967	0.00	0.0157	-3.76	0.4082	-0.29	0.0177	-3.93	0.0024	-1.08		
FAP198A/CYB5D1	TTHERM_01092470	yes	yes	0.8513	-0.09	0.0085	-0.55	0.4064	-0.34	0.0004	-0.98	0.0889	-2.42	0.0710	0.25	0.3138	-0.76	0.2017	-1.37	0.5137	0.84	0.0423	0.20		
FAP198B/CYB5D1	TTHERM_00476710	yes	yes	0.8513	-0.09	0.0927	0.23	0.4064	-0.34	0.4849	0.05	0.0889	-2.42	0.0177	-0.45	0.1449	-1.85	0.2997	-1.01	0.0467	-1.51	0.1430	-0.14		
FAF296	TTHERM_01016220	no	no	nd		0.2675	-0.13	nd		0.4282	-0.06	nd		0.4993	0.08	nd		nd		nd		0.2818	-0.09		
FAF299	TTHERM_00427460	yes	yes	0.8343	0.12	0.2473	0.12	0.5220	0.27	0.0305	0.21	0.6862	0.24	0.0518	0.27	0.3679	-0.38	0.868026	-0.05	0.9871	0.01	0.0982	0.15		

Significant reduction or increase of the protein level are marked in red or blue, respectively; TGD – Tetrahymena Genome Database; LQF – label-free quantification, TMT- tandem mass tag 10-plex isobaric mass tagging approaches.

Protein name potential RS- associated proteins	Acc number in TGD	Co-IP	BioID	RSP3A-KO/WT				RSP3B-KO/WT				RSP3C-KO/WT				CFAP206-KO/WT		CFAP61-KO/WT		CFAP91-KO/WT			
				LQF		TMT		LQF		TMT		LQF		TMT		LQF		LQF		LQF		LQF	
				q-value	diff- erence	q-value	diff- erence	q-value	diff- erence	q-value	diff- erence	q-value	diff- erence	q-value	diff- erence	q-value	diff- erence	q-value	diff- erence	q-value	diff- erence	q-value	diff- erence
TUTPR1	TTHERM_00623040	yes	yes	0.4441	-0.77	0.6006	-0.05	0.1834	-0.66	0.9062	-0.01	0.0091	-3.42	0.0116	-0.56	0.0256	-2.01	0.7405	0.14	0.0683	1.95	0.1189	-0.14
TUTPR2	TTHERM_00221090	yes	yes	0.9944	0.01	0.8607	0.02	0.0192	-2.31	0.0035	-0.34	0.4491	-0.66	0.1096	0.20	0.1426	-1.10	0.3186	-0.52	0.0033	2.22	0.0821	-0.16
kelch-motif containing 1	TTHERM_00760390	yes	yes	0.9112	0.06	0.8595	0.02	0.8452	-0.09	0.1661	0.11	0.9543	-0.04	0.2436	0.14	0.6226	-0.23	0.5444	0.23	0.0053	3.87	0.0017	-0.52
kelch-motif containing 2	TTHERM_00532180	no	yes	0.3220	-1.39	0.8817	0.02	0.0135	-3.18	0.0099	-0.30	0.2992	-1.49	0.8341	-0.02	0.1784	-1.51	0.4491	0.71	0.0646	1.85	0.9241	-0.01
LRR23A	TTHERM_000703669	yes	yes	0.2089	-1.02	0.0663	0.23	0.3478	-0.45	0.0339	0.21	0.2615	-1.73	0.0178	0.42	0.2243	-1.41	0.7830	-0.10	0.0035	4.12	0.0014	-1.16
LRR23B	TTHERM_00105260	yes	yes	0.7001	-0.22	0.6874	0.04	0.3550	-0.40	0.1415	0.12	0.4826	-0.45	0.0131	0.46	0.1661	-0.60	0.0477	-0.94	0.0034	6.02	0.0003	-1.12
LRR24	TTHERM_01220370			0.5447	-0.36	0.9962	0.00	0.4854	-0.30	0.1142	0.13	0.4139	-0.64	0.7251	0.04	0.3822	-0.39	0.5476	0.20	0.7461	0.14	0.1612	0.12
LRR-containing	TTHERM_01084360	no	no	0.5992	0.27	0.0605	0.26	0.1355	-0.95	0.0019	-0.40	0.4602	-0.45	0.0285	0.34	0.0682	-2.21	0.0770	-1.45	0.0085	1.27	0.0132	-0.32
CCDC104FAF36 sperm-tail PG-rich repeat protein 1	TTHERM_00557860	yes	yes	0.6816	0.46	0.0116	-0.72	0.0646	-1.20	0.0006	-1.27	0.6020	0.45	0.0017	-0.77	0.1204	1.10	0.0180	1.68	0.0083	2.01	0.0017	-0.84
sperm-tail PG-rich repeat protein 2	TTHERM_00549640	no	no	0.8430	0.16	0.0267	-0.36	0.0337	-1.43	0.0170	-0.29	0.9056	0.08	0.0044	0.81	0.1105	0.94	0.0546	-1.59	0.0147	3.47	0.0354	-0.90
	TTHERM_00420710	no	no	0.5100	0.64	0.1000	-0.20	0.0149	-0.77	0.0028	-0.38	0.3054	1.02	0.1326	-0.18	0.0186	-3.18	0.1492	-0.64	0.0089	3.72	0.1059	-0.14
EF-hand protein	TTHERM_00637390	no	no	0.4234	-0.75	0.0310	0.31	0.1381	-1.27	0.0278	-0.22	0.5102	-0.62	0.8047	0.03	0.3778	0.83	0.1814	1.43	0.0249	2.21	0.0026	-0.46
EF-hand protein	TTHERM_00642960	yes	yes	0.5110	-0.39	0.8682	0.02	0.2456	-0.48	0.0146	-0.27	0.9841	-0.01	0.3052	0.11	0.9272	-0.05	0.2987	0.43	0.7630	0.17	0.7626	-0.02
EF-hand protein	TTHERM_00193760	yes	yes	0.4058	-0.77	0.5332	-0.06	0.2303	-0.68	0.1101	-0.12	0.3828	-0.76	0.3680	-0.10	0.4723	-0.26	0.6762	0.13	0.0364	0.93	0.0834	0.01

Enzyme name	Acc numer in TGD	Co-IP	BiolID	RSP3A-KO/WT				RSP3B-KO/WT				RSP3C-KO/WT				CFAP206-KO/WT	CFAP61-KO/WT	CFAP91-KO/WT			
ATP homeostasis				LFQ		TMT		LFQ		TMT		LFQ		TMT		LFQ		LFQ		LFQ	
				q-value	diff- rence	q- value	diff- rence	q-value	diff- rence	q-value	diff- rence	q-value	diff- rence	q-value	diff- rence	q-value	diff- rence	q-value	diff- rence	q-value	diff- rence
ENO1	TTHERM_00486480	no	no	0.8338	-0.23	0.0188	0.42	0.0195	-1.96	0.0113	-0.29	0.3673	-1.58	0.5140	0.08	0.0981	-0.80	0.0662	-2.76	0.0045	-3.83
PVK1 (Pyruvate Kinase)	TTHERM_00118600	no	no	0.3864	1.07	0.0089	0.59	0.0526	-1.93	0.0081	-0.31	0.2726	-2.11	0.1047	-0.22	0.1606	-0.72	0.0176	-1.77	0.0042	-6.35
adenylate kinase 1, AK1	TTHERM_00317200	yes	yes	0.2096	-1.03	0.2215	-0.12	0.0186	-1.50	0.0010	-0.49	0.2443	-0.93	0.4282	-0.09	0.3223	-0.43	0.4288	0.29	0.2574	-0.43
adenylate kinase 7A, AK7A	TTHERM_00558060	yes	yes	0.9677	-0.02	0.0706	0.23	0.2455	-0.51	0.6449	0.03	0.9148	-0.06	0.0778	0.24	0.3148	-0.43	0.2183	-0.45	0.0095	-3.57
adenylate kinase 7B, AK7B	TTHERM_000569069	yes	yes	0.6181	-0.27	0.0196	0.33	0.3412	-0.42	0.0116	0.28	0.5422	-0.36	0.0306	0.35	0.3047	-0.42	0.1970	-0.53	0.0048	-6.74
adenylate kinase 8A, AK8A	TTHERM_00227800	yes	yes	0.5095	-0.41	0.4586	0.07	0.0181	-1.80	0.0111	-0.28	0.2103	-1.10	0.6136	-0.05	0.1451	-0.69	0.1838	0.54	0.6095	0.22
adenylate kinase 8B, AK8B	TTHERM_00455540	yes	yes	0.0003	-4.12	0.0065	-0.60	0.1562	-0.73	0.0723	-0.16	0.8016	0.14	0.2832	0.12	0.5638	-0.25	0.5959	0.18	0.2885	0.41
adenylate kinase 9, AK9	TTHERM_00148750	yes	yes	0.4987	-0.40	0.0318	0.30	0.2744	-0.44	0.0031	0.35	0.4566	-0.48	0.0129	0.45	0.4154	-0.30	0.1851	-0.49	0.0034	-5.47
arginine kinase 2, AK2	TTHERM_00151390	no	no	0.3877	-1.80	0.0025	1.28	0.0016	-4.46	0.0009	-0.54	0.0910	-2.16	0.0083	0.66	0.0171	-6.23	0.0107	-5.07	0.0034	-5.39
cGMP-related enzymes																					
guanylate kinase 1	TTHERM_00781030	yes	yes	0.5794	-0.30	0.8152	0.02	0.3717	-0.41	0.0244	0.23	0.5953	-0.33	0.0659	0.25	0.4619	-0.33	0.4632	-0.31	0.0048	-4.83
3'-5'-cyclic nucleotide phosphodiesterase	TTHERM_00293350	no	yes	0.4822	-0.50	0.0101	0.53	0.3325	-0.43	0.0011	0.55	0.5061	-0.49	0.0201	0.75	0.7072	-0.18	0.0107	-4.61	0.0053	-5.50
NDK, nucleoside diphosphate kinase	TTHERM_000160999	no	no	0.3841	1.98	0.0450	0.32	0.2471	-1.28	0.0013	-1.07	0.9015	0.21	0.0136	-0.51	0.9075	0.07	0.2269	-0.56	0.0076	-1.45
Ser/thr kinases																					
caveolin kinase 1, CK1, calcium/calmodulin-dependent protein kinase	TTHERM_00938880	yes	yes	0.3214	-1.20	0.0153	0.43	0.1869	-0.79	0.0009	0.48	0.3032	-0.90	0.0172	0.47	0.3752	-0.42	0.0398	-1.77	0.0040	-3.45
	TTHERM_01344710	no	no	0.9064	-0.07	0.2161	-0.129	0.0317	-1.283	0	-0.88	0.4734	-0.65	0.011	-0.47	0.038	-2.17	0.154	-0.63	0.0075	-4.29
STK36-like	TTHERM_000442899	yes	yes	0.4018	-0.8	0.3671	0.096	0.5252	0.28	0.0028	0.44	0.5808	-0.29	0.6694	0.05	0.0774	1.13	0.0835	0.87	0.0078	1.48
cAMP-dependent kinase catalytic subunit	TTHERM_00433420	yes	yes	0.7840	0.21	0.6428	-0.04	0.0831	-1.36	0.0000	-0.98	0.5403	-0.59	0.0000	-0.61	0.9273	-0.05	0.2436	0.49	0.0206	-1.50
cAMP-dependent kinase catalytic subunit	TTHERM_00658860	no	no	0.5455	-0.68	0.1637	-0.16	0.1845	-1.16	0.0019	-0.63	0.3813	-1.09	0.1141	-0.23	0.2830	0.55	0.3487	0.43	0.0075	-3.58
cAMP-dependent kinase regulatory protein	TTHERM_00623090	yes	yes	0.8422	0.18	0.3634	-0.09	0.0103	-3.57	0.0035	-1.04	0.8085	0.19	0.0120	-0.47	0.0945	-0.96	0.6476	-0.16	0.0073	-3.93

Significant reduction or increase of the protein level are marked in red or blue, respectively; TGD – Tetrahymena Genome Database; LFQ – label-free quantification, TMT- tandem mass tag 10-plex isobaric mass tagging approaches.

Figure 6-figure supplement 3. (continued)

Figure 2-Source data1’. The analyses of the swimming rate of wild-type and RSP3 mutant cells

Figure 2-Source data2’. The analyses of the cilia beating frequency in wild-type and RSP3 mutant cells.

Figure 4-Source data1’. LFQ-based identification of proteins with altered (reduced or elevated) levels in radial spoke mutants.

Figure 4-Source data2’. Tandem Mass Tag (TMT)-based identification of proteins with altered (reduced or elevated) levels in radial spoke mutants.

Figure 6-Source data1’. Mass spectrometry analysis of the ciliary proteins that co-immunoprecipitated with RSP3A-3HA, RSP4A-3HA, RSP4B-3HA, or RSP4C-3HA, all expressed at native levels. The radial spoke proteins are highlighted: RS head (dark blue), RS neck (yellow), RS stalk (green and light green), former CSC complex components and CFAP206 (violet), new RS components identified by [Gui et al., 2021](#) (light brown), new putative RS components (pink), adenylate kinases (dark yellow), other enzymes (orange), and central apparatus components (cyan).

Figure 6-Source data2’. Mass spectrometry analysis of proteins biotinylated in cells expressing RSP3A, RSP3B, RSP3C, RSP4A, RSP4B, RSP4C, CFAP61, CFAP91, or CFAP206 with a C-terminal –HA-BirA* tag under the control of the respective native promoters. The radial spoke proteins are highlighted: RS head (dark blue), RS neck (yellow), RS stalk (green and light green), former CSC complex components and CFAP206 (violet), new RS components identified by [Gui et al., 2021](#) (light brown), new putative RS components (pink), adenylate kinases (dark yellow), other enzymes (orange), central apparatus components (cyan).

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Reviewer #1 (Public Review):

Summary:

Radial spokes (RS) are made of >20 proteins and are believed to be a transducer to coordinate axonemal dyneins to enable the beating motion of motile cilia. While the atomic structure of RS from green algae *Chlamydomonas* and *H. Sapience* has been solved by single particle cryo-EM recently, this work by Bicka et al. provided the atomic structure of RS from ciliate *Tetrahymena*. They identified component proteins of *Tetrahymena* RS, which correspond to those in the atomic structure of *Chlamydomonas* and human RS. These proteins were likely already guessed as RS components, based on sequence similarity, but in this work experimentally identified for the first time. Furthermore, they discovered novel isoforms of RS proteins and characterized them structurally and functionally. RSP3 has three isoforms (A, B, and C). They are distributed specifically in the three radial spokes within the repeating unit as proved by mutant analysis, cryo-EM, and proteomics. By high-speed video microscopy, they proved the essential roles of RSP3B for ciliary beating. These isoforms have never been reported in past works and this demonstrates the novelty of this work.

Strength:

Their discovery of RSP3 isoforms is unexpected and, although it is still not clear why *Tetrahymena* needs to have these isoforms, will evoke future research. The authors characterized the multi-facet aspects of these proteins precisely, structurally by cryo-EM, functionally by waveform and velocity analysis, and in terms of protein networking by co-IP and bioID studies.

Weakness:

While the first half of this manuscript about RSP3 isoforms is very well organized and described (this reviewer still has some advice to make this story convincing and attractive), the later part has room for improvement. Some results are presented in the current manuscript without mentioning figures or tables, for example in "250: The components of the *Tetrahymena* radial spoke stalks" no figure/table is cited. There is also inconsistency - in 327 RSP9 is mentioned as a MORN protein, but in Fig.6 Sup.3 Table.1, it is mentioned as "unknown".

Reviewer #2 (Public Review):

Summary:

Radial spokes are evolutionarily conserved protein complexes that are important for cilia motility. So far, the composition of certain radial spokes was investigated in the algae *Chlamydomonas*, mice, and humans. This work by Bicka et al. investigated the composition of radial spokes in the ciliate *Tetrahymena* by analyzing knockouts and strains that express tagged radial spoke proteins, using mass spectrometry and cryo-electron tomography. While three specific types of radial spokes have been reported thus far, this study suggests that in *Tetrahymena*, there is another layer to the variability in radial spokes. Additionally, many

proteins with predicted enzymatic folds have now been assigned to radial spokes. The comparison of ciliary complexes between species is important to define the basic principles that govern cilia motility, as well as to reveal the differences that enable cilia of various organisms to beat in diverse environments.

Strengths:

The manuscript includes a thorough bioinformatic analysis of radial spoke proteins in *Tetrahymena* and reveals the presence of multiple orthologs to certain algae and mammalian radial spoke proteins. The mass spectrometry analysis and cryo-electron tomography experiments are solid and informative. This work provides a lot of important data and thus, opens the door to resolve the exact composition and structures of radial spokes in *Tetrahymena* and perhaps other species.

Weaknesses:

The assignment of the three RSP3 orthologs to RS1, RS2, and RS3 is based only on missing structures in the knockouts. Although this method is informative, it is not sufficient to draw conclusions regarding the positions of the missing proteins. There are numerous examples where a structure was missing, but the absent protein was localized elsewhere (i.e., absence of central pair protrusions in patients with mutations in radial spoke proteins). To directly demonstrate the position of an RSP3 ortholog in a certain radial spoke, the protein can be labeled with a tag that is visualized in subtomogram averages (as was done in Oda et al., 2014 and other studies). Relying on the data from knockouts alone, the model for radial spoke composition in *Tetrahymena* (Fig. 6) may be incomplete.

The control for the bio-ID experiment was WT cells. Since there are many hits in the experiment, a better control would have been a strain with free BirA, or BirA fused to a protein that is distant from the radial spokes, such as one of the outer-dynein arm proteins, or a ciliary membrane protein.

Reviewer #3 (Public Review):

Summary:

The authors aim to study the role of axoneme radial spoke proteins in forming the three radial spokes that connect the central pair microtubules with the doublet microtubules of the ciliary axoneme. They combined existing and novel mutants to first study ciliary dynamics, followed by cryoET structure and proteomics to identify known and new radial spoke protein components, and assign those with radial spoke(s) to which they belong.

Strengths: / Weaknesses:

The strengths of this study are in the genetic mutants combined with the cryoET to study the unique structural impacts of each mutant on the three radial spokes. The proteomics to study protein loss and interactions also enabled a comprehensive comparison of proteins at the radial spoke under normal and mutant conditions. This allowed the authors to predict that there are several classes of each type of radial spoke. While there are some limitations with overlapping phenotypes between the mutants, this tactic allows the authors to predict known and new proteins that are predicted to localize to each of the three radial spokes. However, in some places, the conclusions are overstated and the list of molecules without functional insight simply identifies new components that will need to be the target of future studies. Two examples of this are that the authors claim to have "solved the composition of individual radial spokes" and that "adenylate kinases [that] dock to specific RSs". Neither of these statements should be made based on the results in this manuscript. Moreover, the authors state that Rsp3Bp does not change in *rsp3C* knockouts and conclude that Rsp3B from the A-C heterodimer is still attached to the axoneme without maintaining the RS2 structure. To me, this makes a series of strongly stated conclusions without the results to justify the statement. The authors also report on unique features of ciliary dynamics resulting from the loss of each of the three *Tetrahymena* RSP3 genes. This showed a strong phenotype for *rsp3b* knockout.

However, a quantitative measure of ciliary dynamics to understand how much the presented data represent the ciliary dynamics was not clear. Furthermore, the authors argue that metachrony or coordination between cilia was affected but the presented data are not interpretable or quantified. Furthermore, the authors state that all three Rsp3 paralogs localize along the entire length of the cilium. However, Rsp3A and B do not localize to ciliary tips, while Rsp3C does. This may inform the differences found in the ciliary waveform for rsp3C mutants compared to rsp3A and rsp3B. The authors state that they have defined a "large part of the protein composition of individual RSs...". It is not clear to me that they know how much of the total RS proteome they have identified.

This manuscript identifies new candidate proteins that may function with radial spokes, future work will be required to 1) confirm their localization to the radial spoke and 2) to study their function within radial spokes.

Author Response

We thank Editors and Reviewers for their positive evaluation of our work and appreciation of new findings and applied interdisciplinary approaches. We also thank for pointing out manuscript weaknesses as well as for all suggestions and advices that can strengthen this manuscript. We apologise for mistakes, overstatements or discrepancies in citing figures as well as omitted references.

The first part of the manuscript focuses on the *Tetrahymena* RSP3 genes mutants. *Tetrahymena* genome encodes three RSP3 paralogs that are the components of different radial spokes and likely form homo- and heterodimers. Thus, the proteomic analyses of *Tetrahymena* radial spokes are more complicated compared to the similar analyses in organisms having a single RSP3 protein.

Next, we attempted to identify proteins specific for each RS type. Conducting this research, we took advantage of six different radial spoke knockout mutants (RSP3A-KO, RSP3B-KO, RSP3C-KO, CFAP206-KO, CFAP61-KO, and CFAP91-KO) and compared wild-type and mutants' ciliomes using two methods, LFQ and TMT (for each mutant the experiment was repeated three times). Comparative analyses of the wild-type and mutants ciliomes allowed us to identify *Tetrahymena* radial spoke proteins, in the case of RS1 (WT versus RSP3A-KO), RS2 (WT versus RSP3B-KO, RSP3C-KO, and CFAP206-KO), and RS3 (wild-type versus CFAP61-KO and CFAP91-KO). The extensive proteomic analyses were combined with detailed bioinformatics studies and co-immunoprecipitation and BioID assays to verify the presence of identified proteins in RS complexes.

Importantly, in the case of RS1 and RS2 spokes, our findings are in agreement with data obtained for *Chlamydomonas* and mammalian radial spokes. Thus, it is very likely, that newly discovered RS1 and RS2 proteins as well as identified *Tetrahymena* RS3 proteins are also true RS subunits.

As an outcome of this part, we propose a model of the RS protein composition in a ciliate *Tetrahymena*. We agree that this model requires further experimental verification (for example by pull-down experiments). However, considering the number of identified proteins, this is a considerable amount of additional work that we would like to publish as separate papers. We would like to add that our current analyses of additional RS3 mutant (that will be published separately) support findings regarding RS3 proteomic composition.

Reviewer 2:

The control for the bio-ID experiment was WT cells. Since there are many hits in the experiment, a better control would have been a strain with free BirA, or BirA fused to a protein that is distant from the radial spokes, such as one of the outer-dynein arm proteins, or a ciliary membrane protein.

The BirA* tag is approximately 30 kDa protein and thus it can be transported to cilia by diffusion. BirA* ligase present throughout the cilia could randomly biotinylate proteins including radial spoke proteins. Thus, expression of the BirA* alone is not the best control. We have performed numerous BioID experiments in which BirA* tag was fused with T/TH subunits (CFAP43, CFAP44, Urbanska et al., 2018), subunits of the small complex positioned parallel to N-DRC (CCDC113, CCDC96, Bazan et al., 2021), CFAP69, SPEF2A (C1b central apparatus complex, Joachimiak et al., 2021), N-DRC proteins (Ghanaeian et al., Biorxiv, 2023) and subunits of other ciliary complexes (our unpublished data). The comparison of the earlier obtained BioID data with RSP BioID data, prove that identified proteins are specifically associated with radial spokes. Therefore, in our model, wild-type cells are a good control for BioID experiments.